6

Spectral Estimation of Random Signals

LEARNING OBJECTIVES

In this chapter, we will learn about the following concepts.

- Autoregressive (AR) model estimation
- Energy density spectrum
- Power density spectrum
- Spectrum estimation using discrete Fourier transform (DFT)
- Bartlett method
- Welch method
- Blackman–Tukey method
- Parametric methods
- Spectral estimation and detection of harmonic signals
- Music algorithm
- Maximum entropy method (MEM)
- Auto regressive moving average (ARMA) model estimation
- Moving average (MA) model estimation
- Minimum variance estimation
- Eigen analysis algorithm for spectrum estimation
- Applications of power spectrum estimation: formant estimation
- Cepstrum evaluation
- Pitch period measurement using cepstrum
- Cumulant estimation
- Bispectrum estimation

In this chapter, we will discuss the classification of signals into two classes, namely, energy signals and power signals. The exact evaluation of the spectrum is not possible. Hence, the spectrum of random signals like speech needs to be estimated. Different methods for spectrum estimation such as the periodogram method, Welch method, and the Blackman–Tukey method will be described in this chapter. We will introduce the signal models AR, MA, and ARMA. Power spectrum estimation using parametric methods such as MA model parameters, AR model parameters, and ARMA model parameters will be

explained. We will also deal with other spectrum evaluation methods like minimum variance method and eigenvalue algorithm method. The cepstrum domain is described, and its use for pitch period measurement of speech signal is illustrated. Finally, higher order spectrum (HOS) estimation called cumulant spectra is described. The evaluation of cumulant spectra and HOS estimation is dealt with in detail.

6.1 Estimation of Density Spectrum

In this section, we will discuss the estimation of density spectrum for two classes of signals—energy and power. Spectrum estimation is not without problems. First, we only have a finite length of data record from which we have to estimate the spectrum; second, we do not know the probability density functions of naturally occurring signals like speech, and electrocardiogram (ECG), electroencephalogram (EEG) signals.

6.1.1 Classification of Signals

Signals can be classified in the following ways.

1. Periodic and aperiodic signals: A periodic signal $x(t)$ is a function for which there exists some value of T that satisfies $x(t+T)=x(t)$ for all t. Here, T denotes the period. For example, consider a sinusoidal signal given by $x(t)=A\sin(t)$. Here, the signal is periodic with a period of 2π. For some signals, no such value of T exists for which $x(t+T)=x(t)$ holds good. Such signals are termed aperiodic signals.
2. Energy and power signals: For a signal $x(t)$, x represents the amplitude or magnitude of the signal. The instantaneous power is given by $|x(t)|^2$. The instantaneous power when integrated over its duration gives the total energy. The generalized format for representing the total energy of a signal $x(t)$ is

$$E=\int_{-\infty}^{\infty}|x(t)|^2\,\mathrm{d}t \tag{6.1}$$

or

$$E=\lim_{T\to\infty}\int_{-T}^{T}|x(t)|^2\,\mathrm{d}t. \tag{6.2}$$

Power is nothing but energy per unit time. To calculate power over a period, say, $2T$, we divide the energy over the period $2T$ by the value $2T$. The average power is calculated as the limit of power over the period by allowing the period to tend to infinity. Mathematically, power can be written as

$$P=\lim_{T\to\infty}\frac{1}{2T}\int_{-T}^{T}|x(t)|^2\,\mathrm{d}t. \tag{6.3}$$

Energy signals have finite energy and zero average power. Power signals have infinite energy but the average power is finite. Signals for which both energy and average power are infinite are neither energy signals nor power signals. An example of such a signal is white noise (idealized form of noise; its power density is independent of operating frequency). The adjective "white" is used as white light contains equal amounts of all frequencies within the visible band of electromagnetic radiation. Periodic signals are power signals. They exist over infinite time and the total energy is infinite. The average power over a period is finite. For example, a sinusoidal signal is a power signal and a short-lived aperiodic signal is an energy signal.

6.1.2 Power Spectral Density and Energy Spectral Density

In this section, energy spectrum and power spectrum are explained. Naturally occurring signals have some random component. We cannot predict the next sample from the previous samples using a mathematical model. Moreover, naturally occurring signals are always of infinite duration and we have only finite duration signals available with us. Using this finite duration signal, we try to predict the characteristics of the signal such as mean and variance.

6.1.2.1 Computation of Energy Density Spectrum of Deterministic Signals

Consider a deterministic signal having finite energy and zero average power. This signal is called an energy signal. We will estimate the energy density spectrum of this energy signal. Let us consider $x(t)$ as a finite duration signal. The problem is to estimate the true spectrum from a finite duration record. If $x(t)$ is a finite energy signal, the energy E of the signal is finite and is given by

$$E = \int_{-\infty}^{\infty} |x(t)|^2 \, \mathrm{d}t < \infty. \tag{6.4}$$

The Fourier transform (FT) exists and is given by

$$X(F) = \int_{-\infty}^{\infty} x(t) \mathrm{e}^{-j2\pi Ft} \, \mathrm{d}t. \tag{6.5}$$

We know that energy is neither created nor destroyed. Hence, the energy of the time domain signal is the same as the energy of the Fourier domain signal. We can write

$$E = \int_{-\infty}^{\infty} |x(t)|^2 \, \mathrm{d}t = \int_{-\infty}^{\infty} |X(F)|^2 \, \mathrm{d}F. \tag{6.6}$$

The symbol $|X(F)|^2$ represents the distribution of signal energy with respect to frequency and is called the energy density spectrum. It is denoted as $S_{XX}(F)$. The total energy is calculated as the integration of $S_{XX}(F)$ over all frequencies F. We will show that $S_{XX}(F)$ is the FT of an autocorrelation function.

The autocorrelation function of the finite energy signal is given by

$$R_{XX}(\tau) = \int_{-\infty}^{\infty} x^*(t) \, x(t+\tau) \mathrm{d}t, \tag{6.7}$$

where $x^*(t)$ is the complex conjugate of $x(t)$. Let us calculate the FT of this autocorrelation.

$$\int_{-\infty}^{\infty} R_{XX}(\tau)e^{-j2\pi F\tau}\,d\tau = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x^*(t)\,x(t+\tau)e^{-j2\pi F\tau}\,dt\,d\tau. \tag{6.8}$$

Put $t' = t + \tau$, $dt' = d\tau$.

$$\int_{-\infty}^{\infty} x^*(t)e^{j2\pi Ft}\,dt \int_{-\infty}^{\infty} x(t')e^{-j2\pi Ft'}\,dt' = X^{*(F)}X(F) = |X(F)|^2 = S_{XX}(F)\,. \tag{6.9}$$

Hence, $R_{XX}(\tau)$ and $S_{XX}(F)$ are FT pairs.

We can prove the result for discrete time (DT) signals as well. Let $r_{XX}(k)$ and $S_{XX}(f)$ denote the autocorrelation and energy density for a DT signal.

$$r_{XX}(k) = \sum_{n=-\infty}^{\infty} x^*(n)\,x(n+k). \tag{6.10}$$

Put $n + k = n'$. The FT of the autocorrelation is given by

$$S_{XX}(f) = \sum_{k=-\infty}^{\infty} r_{XX}(k)e^{-j2\pi kf} = \sum_{n=-\infty}^{\infty} x^*(n)e^{j2\pi nf}\sum_{k=-\infty}^{\infty} x(n')e^{-j2\pi n'f} = |X(f)|^2. \tag{6.11}$$

We will list two different methods for computing energy density spectrum, namely, the direct method and the indirect method.

Direct method: In the direct method, one computes the FT of the sequence $x(n)$ and then calculates the energy density spectrum as

$$S_{xx}(f) = |X(F)|^2 = \left|\sum_{n=-\infty}^{\infty} x(n)e^{-j2\pi fn}\right|^2. \tag{6.12}$$

Indirect method: In the indirect method, the autocorrelation is computed first. Then, the FT is found to get the energy density spectrum.

$$S_{xx}(f) = \sum_{k=-\infty}^{\infty} \gamma_{XX}(k)e^{-j2\pi kf}. \tag{6.13}$$

There is a practical problem involved in the computation of the energy density spectrum using Equations 6.12 and 6.13. The sequence is available only for a finite duration. Availability of only a finite sequence is equivalent to multiplication of the sequence by a rectangular window. This results in spectral leakage due to Gibbs phenomenon as discussed in Chapter 5.

6.1.2.2 Estimation of Power Density Spectrum of Random Signals

Stationary random signals are a class of signals of infinite duration and infinite energy. The average power is finite. Such signals are classified as power signals. We will estimate the power density spectrum for such signals. For a random process, the autocorrelation is given by

$$\gamma_{XX}(\tau) = E[x^*(t)\,x(t+\tau)]\,, \tag{6.14}$$

where E denotes expected value. This cannot be evaluated if the probability density function (pdf) is not known. For a random process, pdf is not known. We can only estimate the value of autocorrelation as

1. We have records of only finite duration.
2. pdf is not known.

We assume that the process is wide sense stationary and also ergodic. We calculate the time average autocorrelation and equate it to ensemble autocorrelation in case of an ergodic process. Hence, we calculate

$$\gamma_{XX}(\tau) = \lim_{\tau\to\infty}[R_{XX}(\tau)]\,. \tag{6.15}$$

The estimate $P_{XX}(F)$ of the power density spectrum is calculated as the FT of autocorrelation, and the actual value of the power density spectrum is then the expected value of $P_{XX}(F)$ in the limit as $T_0 \to \infty$.

$$\begin{aligned} P_{XX}(F) &= \int_{-T_0}^{T_0} R_{XX}(\tau)\mathrm{e}^{-j2\pi F\tau}\,\mathrm{d}\tau \\ &= \frac{1}{2T_0}\int_{-T_0}^{T_0}\left[\int_{-T_0}^{T_0} x^*(t)x(t+\tau)\,\mathrm{d}t\right]\mathrm{e}^{-j2\pi F\tau}\,\mathrm{d}\tau \\ &= \frac{1}{2T_0}\left|\int_{-T_0}^{T_0} x(t)\mathrm{e}^{-j2\pi Ft}\,\mathrm{d}t\right|^2. \end{aligned} \tag{6.16}$$

The actual power density spectrum is obtained as

$$\Gamma_{XX}(F) = \lim_{T_0\to\infty} E(P_{XX}(F))\,. \tag{6.17}$$

There are two approaches for computation of power density spectrum, namely, the direct method as given by Equation 6.14 or the indirect method by first calculating the autocorrelation and then finding the FT of it to get the power density spectrum.

In practice, we have only finite samples of the signals. The time average autocorrelation is computed using the equation

$$\gamma_{XX}(m) = \frac{1}{n - |m|} \sum_{n=|m|}^{N-1} x^*(n)\, x(n+m) \quad m = -1, -2, \ldots, 1 - N. \tag{6.18}$$

The normalization factor results in a mean value of autocorrelation, which is computed as autocorrelation. We find the FT of the mean value of autocorrelation given by Equation 6.16 to get an estimate of the power spectrum, which can also be written as

$$P_{XX}(f) = \frac{1}{N} \left| \sum_{n=0}^{N-1} x(n)\, \mathrm{e}^{-j2\pi fn} \right|^2 = \frac{1}{N} |X(f)|^2. \tag{6.19}$$

Equation 6.17 represents the well-known form of the power density spectrum estimate and is termed as periodogram. To find the pdf, one draws the histogram for the signal with finite duration. Then, curve fitting tests such as chi-square test or Kolmogorov–Smirnov (K–S test) are used to fit a standard probability distribution for the given signal. The mathematical equation for that standard pdf will be used in calculations.

Concept Check

- What is energy density spectrum? Why is it estimated?
- What are power signals?
- How do we estimate power density spectrum?
- What is the relation between autocorrelation and power density spectrum?
- How will you fit a standard pdf to a random signal?

6.2 Nonparametric Methods

Nonparametric methods are also used to estimate the power spectrum. These methods do not make any assumption about how the data are generated. To be practicable, the methods use finite duration data record, and its use is equivalent to assuming that the data pass via a window of length, say, *N*. The spectral resolution is limited to 1/*N*. However, the limitation on spectral resolution is compensated by the smaller variance for the estimate achieved by these methods. Later in this chapter, we will realize that the product of spectral resolution and variance of the estimate is constant.

We will now discuss power spectral estimation using discrete Fourier transform (DFT). We will also describe different periodogram methods such as the Bartlett method, the Welch method, and the Blackman–Tukey method.

6.2.1 Use of DFT for Power Spectrum Estimation

In this section, we will discuss the basic method of estimating the power spectrum, namely, using DFT. Consider N data samples. We list the algorithm to find the samples of the periodogram as follows.

1. Read N data values.
2. Find N point DFT.
3. The computation of DFT gives N points of the periodogram.

$$P_{XX}\left(\frac{k}{N}\right)=\frac{1}{N}\left|\sum_{n=0}^{N-1}x(n)\mathrm{e}^{-j2\pi nk/N}\right|^2 \text{ for } k=0,1,...,N-1. \tag{6.20}$$

4. Increase the length of the sequence by padding extra zeros to interpolate the spectrum.

The block schematic for using DFT to find the periodogram is shown in Figure 6.1. Speech samples (N) are taken, DFT is calculated, and N points of the periodogram are found. We will then increase the length of the sequence by padding zeros and observe the effect on the plot of the periodogram. Figure 6.2 shows the plot of speech signal samples and the plot of its periodogram. If we pad zeros to the signal, the effect of padding zeros on the periodogram is shown in Figure 6.3. We can observe that the new periodogram is only an interpolation of the earlier periodogram. The spectral resolution does not change. The frequency resolution will depend only on the length of the data samples.

```
%a program to find periodogram using DFT
clear all;
fp=fopen('watermark.wav');
fseek(fp,224000,-1);
a=fread(fp,1024);
a=a-128;
subplot(2,1,1);plot(a);title('plot of voiced part of a signal');
xlabel('sample no.');ylabel('amplitude');
%1024 samples of the voiced signal are displayed
b=fft(a);
b1=abs(b.*b);
subplot(2,1,2);plot(b1);
title('plot of periodogram using DFT');
xlabel('sample no.');ylabel('squared amplitude');
for i=1:2048,
    if i<1025,
        a1(i)=a(i);
```

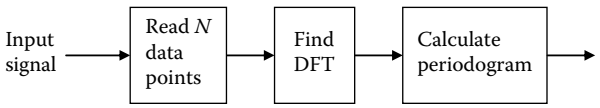


FIGURE 6.1
Block schematic for finding a signal's periodogram.

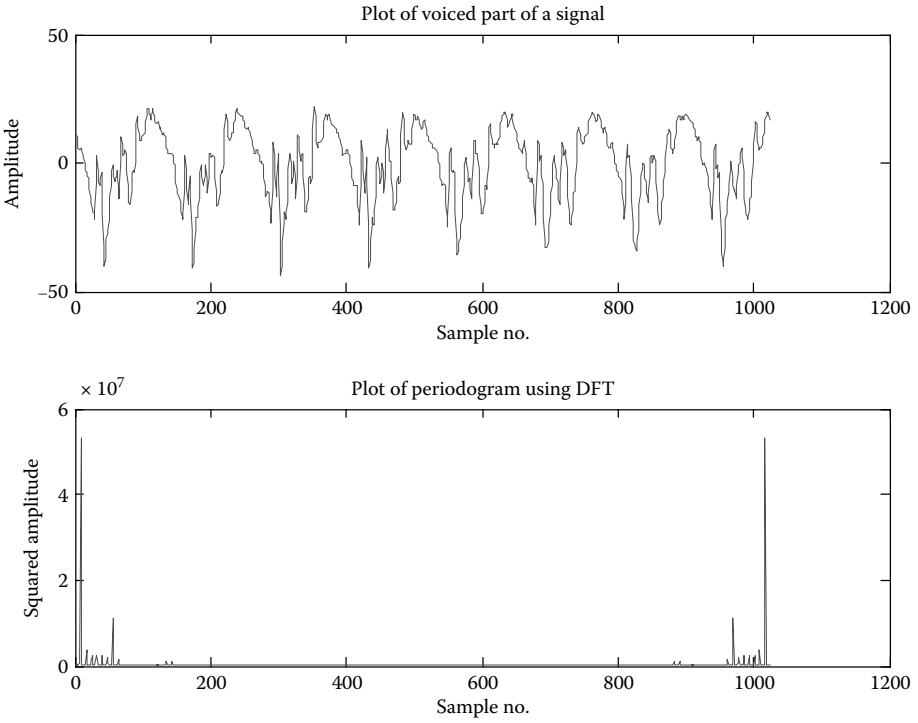


FIGURE 6.2
Plot of signal and its periodogram.

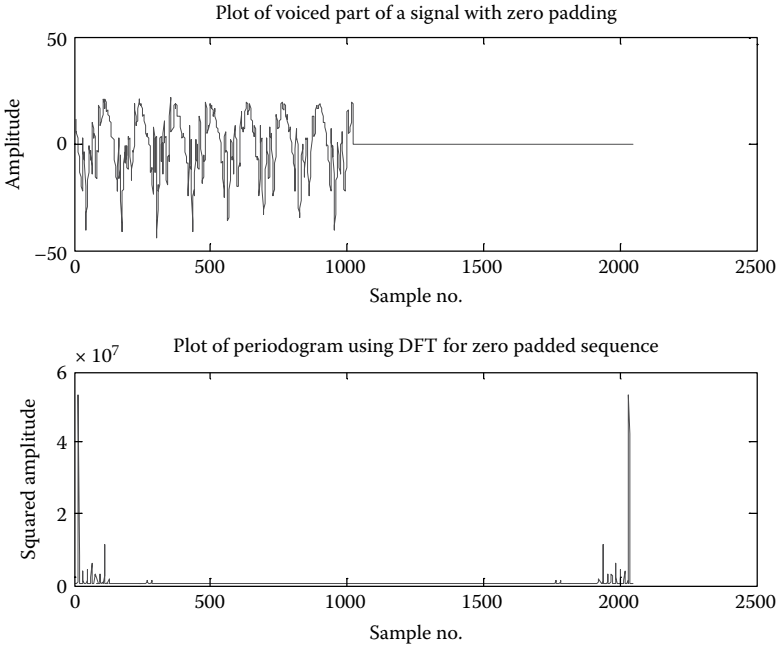


FIGURE 6.3
Plot of signal and its periodogram after zero padding.

```
    else
        a1(i)=0;
    end
end
    figure;
 subplot(2,1,1);plot(a1);title('plot of voiced part of a signal with
zero padding');
xlabel('sample no.');ylabel('amplitude');
b=fft(a1);
b1=abs(b.*b);
subplot(2,1,2);plot(b1);
title('plot of periodogram using DFT of zero padded sequence');
xlabel('sample no.');ylabel('squared amplitude');
```

6.2.2 Bartlett Method

The Bartlett method or the method of averaging periodograms reduces the variance of the power density spectrum estimate. It involves the following three steps.

1. Read N points of data. Divide it in K nonoverlapping segments each of length M.
2. Compute the periodogram for each segment.

$$P_{XX}{}^{i}\left(\frac{k}{M}\right)=\frac{1}{M}\left|\sum_{n=0}^{N-1}x(n)\mathrm{e}^{-j2\pi nk/M}\right|^2 \text{ for } k=0,1,...,N-1. \tag{6.21}$$

3. Lastly, average the periodograms for the k segments to get the Bartlett power spectrum estimate.

$$P_{XX}^{B}\left(\frac{k}{M}\right)=\frac{1}{K}\sum_{i=0}^{K-1}P_{XX}^{i}\left(\frac{k}{M}\right). \tag{6.22}$$

The block schematic for the Bartlett method is shown in Figure 6.4. It is observed that the true spectrum convolves with the frequency characteristic of Bartlett window; hence, the name Bartlett method. The data length is effectively reduced by a factor K. The spectral width increases by the same factor K, and it reduces the frequency resolution. The reduction in frequency resolution results in reduced variance of the power density spectrum estimate by the same factor K. The result of execution of the following MATLAB program is shown in Figure 6.5. The figure is a plot of the signal and its periodogram using the Bartlett method.

```
%a program to find periodogram using Bartlett method
clear all;
fp=fopen('watermark.wav');
fseek(fp,224000,-1);
a=fread(fp,1024);
a=a-128;
subplot(2,1,1);plot(a);title('plot of voiced part of a signal');
xlabel('sample no.');ylabel('amplitude');
```

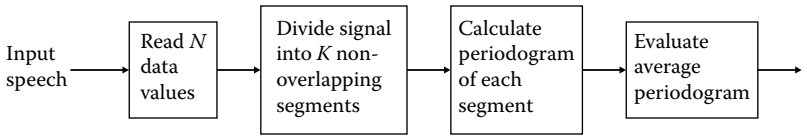


FIGURE 6.4
Block schematic for finding the periodogram using the Bartlett method.

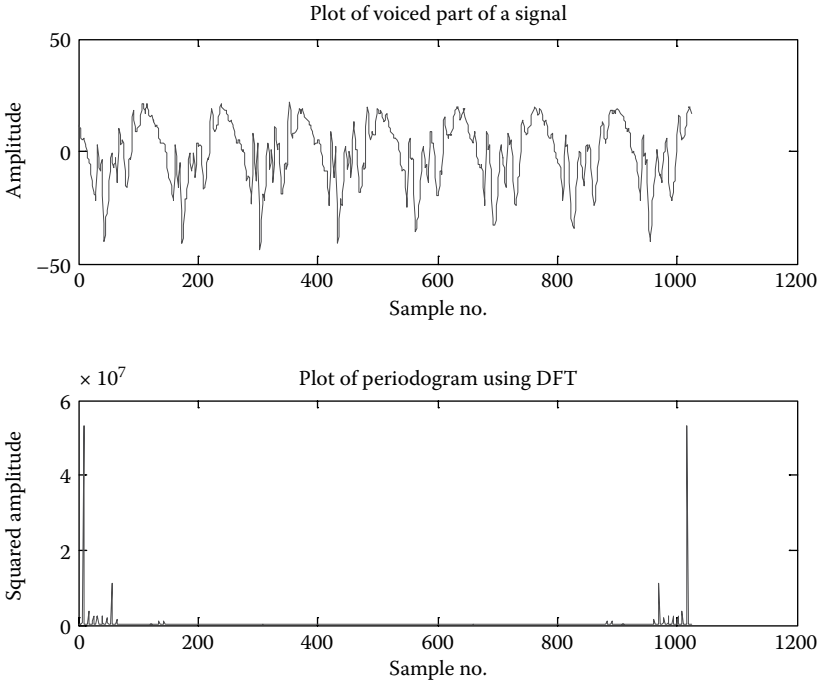


FIGURE 6.5
Plot of signal and its periodogram using the Bartlett method.

```
%1024 samples of the voiced signal are displayed
for i=1:256,
    a1(i)=a(i);
end
for i=257:512,
    a2(i-256)=a(i);
end
for i=513:1024,
    a3(i-512)=a(i);
end
for i=1025:1536,
    a4(i-1024)=a(i);
end
b1=abs(fft(a1));
b2=abs(fft(a2));
b3=abs(fft(a3));
b4=abs(fft(a4));
b11=abs(b1.*b1);
```

```
b22=abs(b2.*b2);
b33=abs(b3.*b3);
b44=abs(b4.*b4);
for i=1:256,
    b(i)=(b11(i)+b22(i)+b33(i)+b44(i))/4;
end
subplot(2,1,2);plot(b);
title('plot of periodogram using Bartlett method');
xlabel('sample no.');ylabel('squared amplitude');
```

6.2.3 Welch Method

This method is also called the method of modified averaging periodogram. The Welch method has two basic modifications on the Bartlett method. It allows the segments to overlap. The overlap used is usually 50%. The use of overlap preserves the statistical properties of the signal. The second modification suggested by Welch is to window the segments prior to computing the periodograms. When a segment passes via a window, the time resolution and frequency resolution are fixed according to the Heisenberg uncertainty principle. If the window length reduces, time resolution improves at the cost of the frequency resolution. It is because of these modifications that the method is also termed the method of modified periodograms. The power spectrum is normalized by a factor equal to the window energy. The Welch estimate is calculated as the average of the periodograms for all segments. As mentioned earlier, the method is at the cost of the frequency resolution but reduces the variance of the power spectrum estimate. The algorithm can be stated as follows.

1. Read the data values. Divide the data segment into L overlapping segments (say, 50% overlap)
2. Pass each of the data segment via a window.
3. Find the periodogram for each segment and divide it by the window energy.
4. Find the average periodogram.

A block schematic for finding the periodogram using the Welch method is shown in Figure 6.6.

A MATLAB program to find the periodogram using the Welch method is given here. We have used a Bartlett window for windowing each segment. Figure 6.7 shows a plot of a signal and its periodogram using the Welch method.

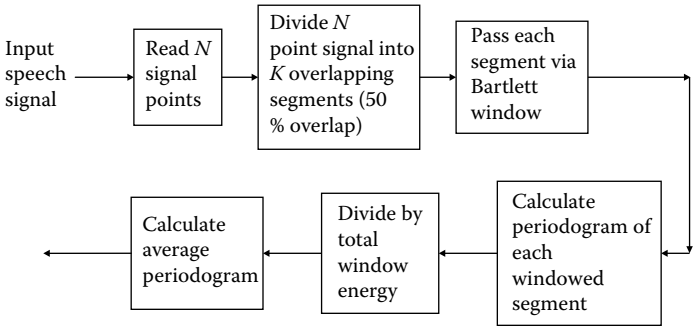


FIGURE 6.6
Block schematic for finding periodogram using the Welch method.

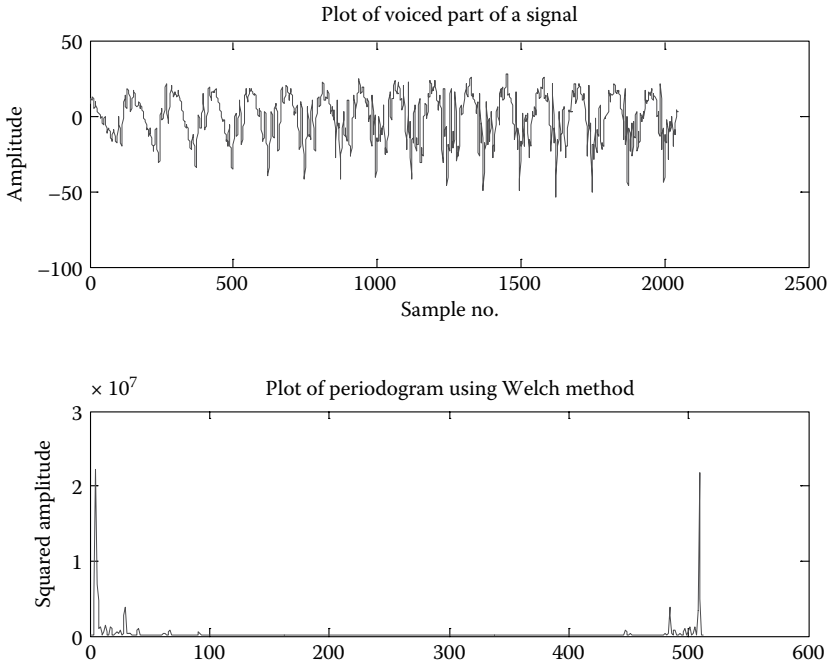


FIGURE 6.7
Plot of signal and periodogram using the Welch method.

```
%a program to find periodogram using Welch method
clear all;
fp=fopen('watermark.wav');
fseek(fp,228000,-1);
a=fread(fp,2048);
a=a-128;
subplot(2,1,1);plot(a);title('plot of voiced part of a signal');
xlabel('sample no.');ylabel('amplitude');
%Divide the data in 50 % overlapping segments and pass it via a window
w=window(@bartlett,512);% Bartlett window function
for i=1:512,
    a1(i)=a(i)*w(i);
end
for i=257:768,
    a2(i-256)=a(i)*w(i-256);
end
for i=513:1024,
    a3(i-512)=a(i)*w(i-512);
end
for i=769:1280,
    a4(i-768)=a(i)*w(i-768);
end
for i=1025:1536,
    a5(i-1024)=a(i)*w(i-1024);
end
for i=1281:1792,
    a6(i-1080)=a(i)*w(i-1280);
```

```
end
for i=1537:2048,
    a7(i-1536)=a(i)*w(i-1536);
end
b1=abs(fft(a1));
b2=abs(fft(a2));
b3=abs(fft(a3));
b4=abs(fft(a4));
b5=abs(fft(a5));
b6=abs(fft(a6));
b7=abs(fft(a7));
b11=abs(b1.*b1);
b22=abs(b2.*b2);
b33=abs(b3.*b3);
b44=abs(b4.*b4);
b55=abs(b*b5);
b66=abs(b6.*b6);
b77=abs(b7.*b7);
W=0;
for i=1:512,
    W=W+w(i);%calculation of window energy for normalization
end
for i=1:512,
    b(i)=(b11(i)+b22(i)+b33(i)+b44(i))+b55(i)+b66(i)+b77(i)/W*7;
end
subplot(2,1,2);plot(b);
title('plot of periodogram using Welch method');
xlabel('sample no.');ylabel('squared amplitude');
```

6.2.4 Blackman–Tukey Method

The Blackman–Tukey method uses the method of autocorrelation to find the power spectrum. It passes the autocorrelation via a smooth window to compensate error while calculating autocorrelation. When a lag is large, the estimation of autocorrelation is less reliable. This happens because a smaller number of data points enters into the estimate when the lag is large. The algorithm can be stated as follows.

1. Read the data sequence.
2. Find the autocorrelation of the data sequence.
3. Pass the autocorrelation via a window, say a Bartlett window.
4. Find the FT of the windowed autocorrelation.

A block schematic for finding the periodogram using the Welch method is shown in Figure 6.8.

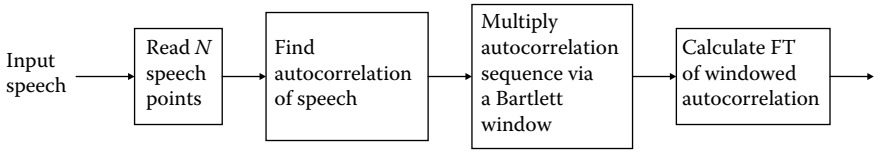


FIGURE 6.8
Block schematic for finding periodogram using the Blackman–Tukey method.

A MATLAB program to find the power spectrum estimate using the Blackman–Tukey method is given here.

```
%to find power spectrum using Blackman-Tukey method
%to find autocorrelation of the signal
clear all;
fp=fopen('watermark.wav');
fseek(fp,228500,-1);
a=fread(fp,2100);
subplot(2,1,1);plot(a);title('plot of voiced part of a signal');
xlabel('sample no.');ylabel('amplitude');

a=a-128;
for k=1:2048,
sum(k)=0;
end
for k=1:2048,
for i=1:45,
sum(k)=sum(k)+(a(i)*a(i+k));
sum(k)=sum(k)/45;
end
end
subplot(2,1,2);plot(sum);title('plot of autocorrelation of a signal');
xlabel('sample no.');ylabel('correlation');
w=window(@bartlett,2048);% Bartlett window function
for i=1:2048,
    sum1(i)=sum(i)*w(i);
end
b=abs(fft(sum1));
figure;
plot(b);title('plot of power density spectrum of a signal');
xlabel('sample no.');ylabel('power density');
```

Figure 6.9 shows a plot of a signal and its autocorrelation and Figure 6.10 shows a plot of FT of the windowed autocorrelation. This is the periodogram using the Blackman–Tukey method.

6.2.5 Performance Comparison of Nonparametric Methods

To compare the performance of the nonparametric methods for estimating power spectrum, the following parameters are used.

1. *Quality*: It is defined as the ratio of the square of mean of the power spectrum estimate and its variance.
2. *Variability*: The reciprocal of quality is defined as the variability. It is the ratio of the variance and the square of the mean of the power spectrum estimate.
3. *Number of computations*: It is assumed that the number of data points N is fixed; the frequency resolution is also specified. The number of complex multiplications required to compute the power spectrum estimate is counted.

The comparison of the Bartlett method, Welch method, and Blackman–Tukey method in terms of quality and number of computations is given in Table 6.1.

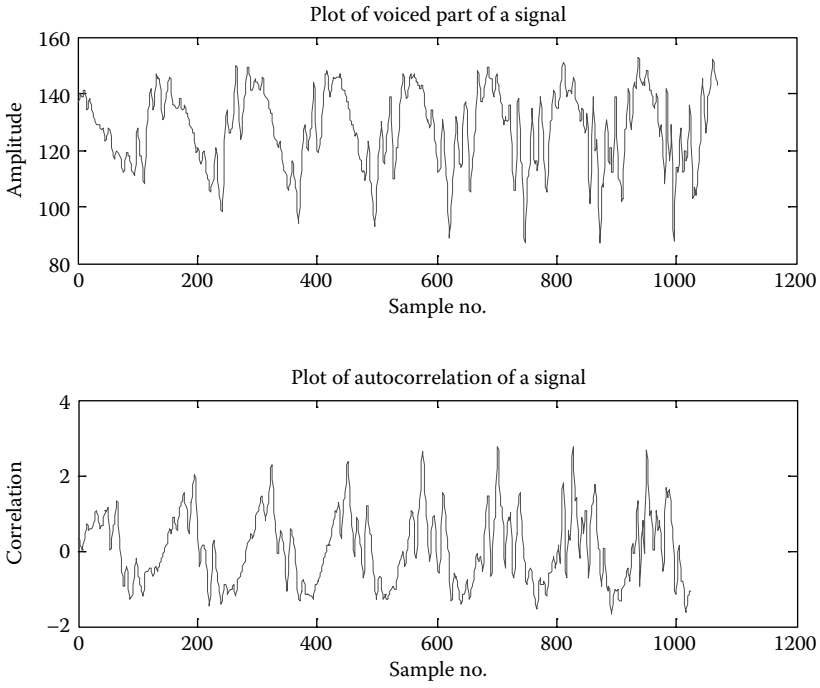


FIGURE 6.9
Plot of signal and its autocorrelation.

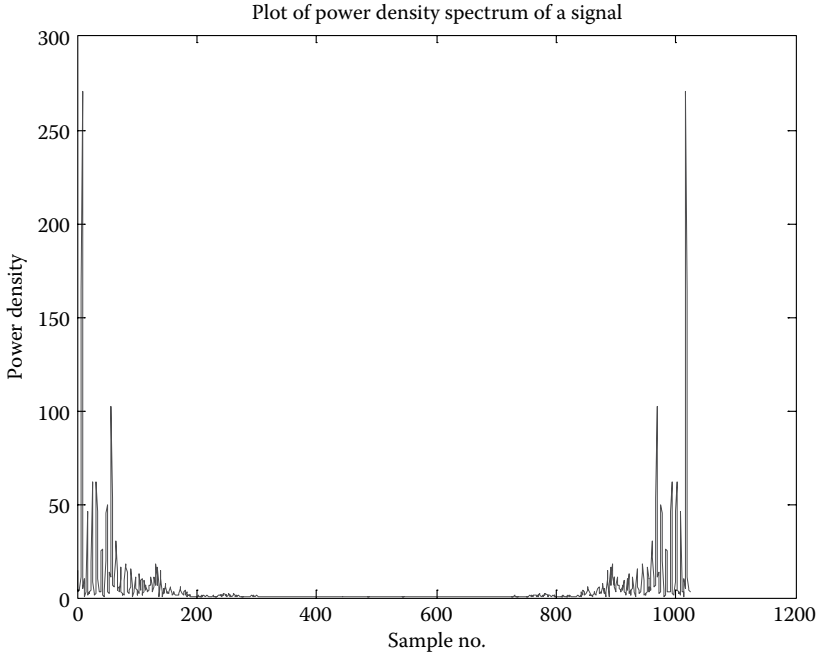


FIGURE 6.10
Plot of FT of windowed autocorrelation (periodogram).

TABLE 6.1
Quality and Number of Computations for Three Nonparametric Methods

Method	Quality	Number of Complex Multiplications
Bartlett method	$1.11N\Delta f$	$\frac{N}{2}\log_2\frac{0.9}{\Delta f}$
Welch method	$1.39N\Delta f$	$N\log_2\frac{5.12}{\Delta f}$
Blackman–Tukey method	$3.34N\Delta f$	$N\log_2\frac{1.28}{\Delta f}$

Referring to Table 6.1, we can see that the Welch method requires little more computational cost than the other methods. Quality is highest for the Blackman–Tukey method. The quality increases as N, the number of points in a sequence, increases. For a given quality, the frequency resolution can be improved by increasing the data length N of a sequence.

Concept Check

- What is a periodogram?
- How will you use DFT for power spectrum calculation?
- What is the method of averaging periodogram?
- Why is 50% overlap used in the Welch method?
- Why is frequency resolution poor in case of the Welch method?
- What is the main advantage of the Welch method?
- What is the difference between the Welch method and the Blackman–Tukey method?
- Name the parameters used for comparing nonparametric methods.
- Which method performs the best?
- Which method uses autocorrelation for computation of periodograms?

6.3 Parametric Methods

Nonparametric models have two basic limitations. First, the autocorrelation estimate is assumed to be zero outside the signal, that is, for $m > N$. The second assumption that limits these methods is that the signal is periodic with a period of N samples. Both these assumptions are nonrealistic. Parametric methods use no such assumptions. They extrapolate the autocorrelation. Extrapolation is possible because the signal has some predictability.

Parametric methods depend on the process of signal generation. Consider a speech signal. We know that there is high correlation between successive speech samples. Further samples can be predicted if previous samples are available. This indicates that the limitation of a finite data record does not impede the parametric approach. We can predict further samples when we have already calculated the predictor coefficients. We have discussed linear prediction in detail in Chapter 5. Lattice method and autocorrelation methods can be used to compute predictor coefficients using the Levinson–Durbin algorithm, Burg algorithm, etc. We have also seen how to calculate the prediction error. The reader may go through these concepts again before going through the parametric methods. Parametric methods include AR model parameters, MA model parameters, and ARMA model parameters. AR model is the most commonly used model.

6.3.1 Power Spectrum Estimation Using AR Model Parameters

The AR model is the most commonly used model for signals like speech. An AR system has only poles, and zeros exist only at zero. It is also called an all-pole model. We will study how the model parameters are used for computation of the power spectrum. Parametric methods are based on modeling signal data sequence as the output of a linear system characterized by a transfer function of the form

$$H(Z) = \frac{\sum_{k=0}^{M} b_k Z^{-k}}{1 + \sum_{k=1}^{N} a_k Z^{-k}}. \tag{6.23}$$

The power density spectrum of the data sequence can be written as

$$\Gamma_{XX}(f) = |H(f)|^2 \, \Gamma_{WW}(f). \tag{6.24}$$

If we assume the input sequence is a zero mean white noise sequence, the autocorrelation is given by

$$\gamma_{WW}(n) = \sigma_W^2 \delta(n). \tag{6.25}$$

$$\Gamma_{XX}(f) = |H(f)|^2 \, \sigma_W^2. \tag{6.26}$$

The Yule–Walker method estimates the autocorrelation from data values and uses the estimate to find the AR model parameters. We have discussed the Levinson–Durbin algorithm for computation of AR model parameters in Chapter 5. The power density spectrum can be estimated as

$$P_{XX}^{YW}(f) = \frac{\sigma_{WP}^2}{\left|1 + \sum_{k=1}^{P} a(k) \mathrm{e}^{-j2\pi fk}\right|^2}. \tag{6.27}$$

P is the order of the predictor and $\sigma_{WP}^2 = E_P^f$ is the forward error of the predictor. The algorithm for power spectrum estimation can be written as follows.

1. Divide the input data sequence into segments of size, say, 128 samples each.
2. Estimate the autocorrelation from the data.
3. Use the Yule–Walker method to find the predictor coefficients.
4. Find the forward error for the predictor.
5. Use Equation 6.25 to evaluate the power spectrum estimation.

The selection of the order of the predictor is important. The AR method of estimation of power spectrum is a block processing one. This means that the method calculates the AR process parameters for a block and estimates its power density spectrum. It then uses the next block for processing. The method may also change or update the coefficients as the next sample arrives. Let us estimate the power density spectrum using the Yule–Walker method. We have used a direct command in MATLAB. A MATLAB program is given here. Figure 6.11 shows a plot of a signal and its power spectrum estimation using the Yule–Walker method.

```
%to find power spectrum using Yule-Walker method.
 clear all;
fp=fopen('watermark.wav');
fseek(fp,224000,-1);
a=fread(fp,1024);
a=a-128;
subplot(2,1,1);plot(a);title('plot of voiced part of a signal');
xlabel('sample no.');ylabel('amplitude');
%1024 samples of the voiced signal are displayed
subplot(2,1,2);
Fs = 22100;   t = 0:1/Fs:.296;
h=spectrum.yulear
d = psd(h,a,'Fs',Fs);
%Calculate the power spectrum estimate using Yule-Walker method
plot(d);
```

We can use the Levinson–Durbin algorithm to find the linear predictor coefficients (LPC) and use a random signal as input to the AR system and find its output. We can plot the power spectrum estimate using the command pyulear in MATLAB.

6.3.2 Burg's Method for Power Spectrum Estimation (Maximum Entropy Method—MEM)

The algorithm uses forward and backward prediction errors and minimizes the sum of squared errors. The LPC can be predicted using Burg's algorithm and the power spectrum can be found using the LPC obtained based on the AR model, same as in Section 6.3.1. The input data samples are from the random process of speech signal, and hence, the pdf for the signal is not known. The process of calculating the power spectrum estimate involves evaluating the FT of autocorrelation. The estimation of autocorrelation can be accurately done if the pdf is known. Normally, the histogram of the signal is compared with known standard density functions using the K–S test, as discussed in Chapter 1, to get the pdf for the unknown data samples. If the exact fit for the known density function is obtained,

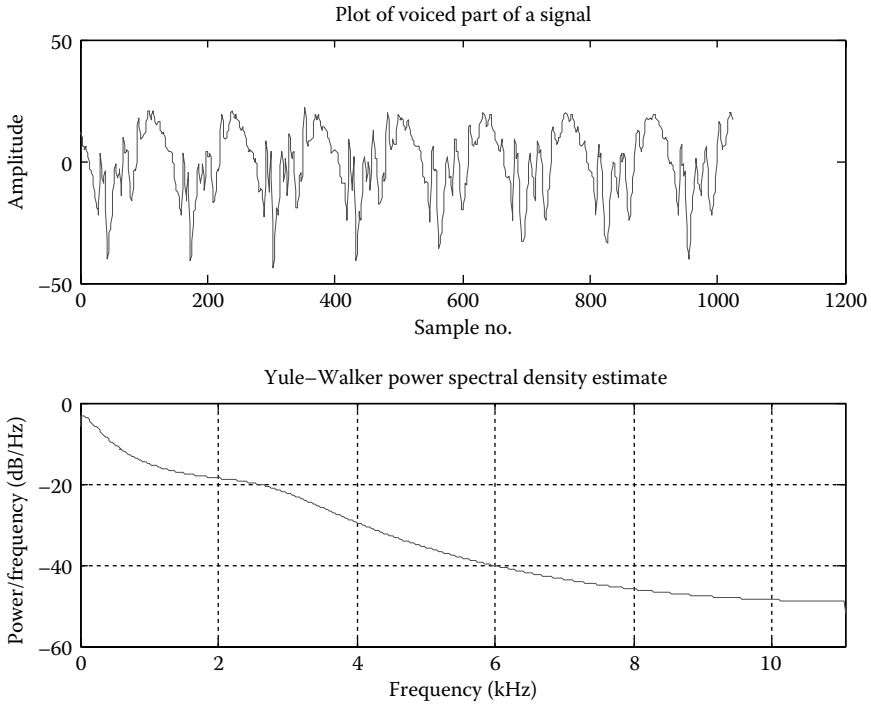


FIGURE 6.11
Plot of signal and its power spectrum using the Yule–Walker method.

which is a very rare situation, then the entropy of the signal is maximized. The power spectrum estimation obtained will then be termed the maximum entropy power spectrum estimation.

Autocorrelation calculation is done for finite data samples. The autocorrelation is then interpolated to find further data samples, which were initially not available. Burg postulated that the interpolation that gives the flattest spectrum will result in the maximum entropy. This is due to the fact that obtaining a flat spectrum is a rare situation and hence has minimum probability of occurrence, which confirms that information is maximized for such a situation. The method of power spectrum estimation based on this postulation is called maximum entropy method (MEM) for power spectrum estimation.

Burg's method uses the lattice filter structure for the system and evaluates reflection coefficients using forward and backward prediction. The power spectrum estimation can be found as follows.

$$P_{XX}^{B}(f) = \frac{E_P}{\left|1 + \sum_{k=1}^{P} a(k)\mathrm{e}^{-j2\pi fk}\right|^2}. \tag{6.28}$$

$$E_P = \sum_{n=P}^{N-1} |f_P(n)|^2 + |g_P(n)|^2 \text{ forward \& backward error.}$$

Burg's method results in a stable filter, gives high resolution, and is found to be computationally efficient. However, it exhibits line splitting at high signal-to-noise ratio. The actual single spectral line shows two closely spaced peaks. In case of higher order models, it shows spurious peaks. For a sinusoidal signal, the spectrum is sensitive to the initial phase. Modifications, namely, windowing and weighting, are found to reduce the drawbacks.

The MATLAB command pburg finds the power spectral density (PSD) estimate using Burg's method. The frequency is expressed in units of radians/sample. ORDER is the order of the autoregressive (AR) model that is used to produce the PSD. pburg uses a default fast fourier transform (FFT) length of 256, which in turn determines the length of P_{XX}. A MATLAB program to find PSD using Burg's method is given here. Figure 6.12 shows a plot of the signal and its PSD.

```
clear all;
fp=fopen('watermark.wav');
fseek(fp,224000,-1);
a=fread(fp,1024);
a=a-128;
subplot(2,1,1);plot(a);title('plot of voiced part of a signal');
xlabel('sample no.');ylabel('amplitude');
%1024 samples of the voiced signal are displayed
subplot(2,1,2);
Fs = 22100;   t = 0:1/Fs:.296;
[p t]=pburg(a,12);
%Calculate the power spectrum estimate using Burg's method
plot(p); title('plot of PSD using Burg's method');
xlabel('sampling time');ylabel('amplitude');
```

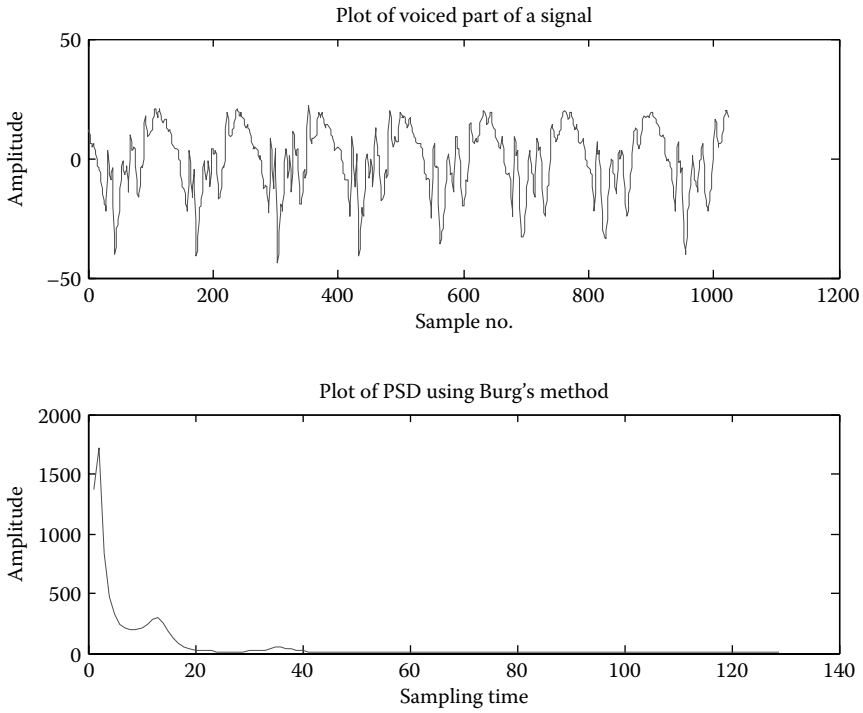


FIGURE 6.12
Plot of signal and its PSD using Burg's method.

6.3.3 Power Spectrum Estimation Using ARMA Model

The ARMA model has poles as well as zeros. It improves the power spectrum estimate using less number of coefficients. This model is suitable when the signal is corrupted by noise. If the signal is generated using AR model and it has additive noise, the overall system will become an ARMA model. The model parameters can be estimated using the least squares technique. The estimated power spectrum can be written as

$$P_{XX}^{\text{ARMA}}(f) = \frac{P_{VV}^{MA}(f)}{\left|1 + \sum_{k=1}^{P} a(k) e^{-j2\pi fk}\right|^2}.$$

$$P_{VV}^{\text{MA}}(f) = \sum_{m=-q}^{q} r_{VV}(m) \mathrm{e}^{-j2\pi fm}.$$

$\gamma_{VV}(m)$: an estimate of autocorrelation for the MA model. (6.29)

6.3.4 Power Spectrum Estimation Using MA Model

The power spectrum for an MA model can be estimated using MA model parameters, which represent parameters of an all-zero model. The power spectrum can be written as

$$P_{XX}^{MA}(f) = \sum_{m=-q}^{q} r_{XX}(m) \mathrm{e}^{-j2\pi fm}. \tag{6.30}$$

$\gamma_{XX}(m)$: an estimate of autocorrelation for the MA model.

The parameters of the MA model can be determined using least squares inverse design.

Concept Check

- List some methods to find autocorrelation coefficients.
- How will you use AR parameters for estimating the power spectrum?
- If the signal is corrupted with noise, which model will you use for power spectrum estimation

6.4 Other Spectral Estimation Methods

This section describes other methods for power spectrum estimation, namely, minimum variance spectral estimation and eigen analysis algorithm for power spectrum estimation.

6.4.1 Minimum Variance Power Spectrum Estimation

Consider an finite impulse response (FIR) filter with coefficients a_k $0 \le k \le P$. We will not constrain a_0 to be unity. Consider an input data sequence $x(n)$. This data sequence is passed via this FIR filter. The output of the filter can be written as

$$y(n) = \sum_{k=0}^{P} a_k x(n-k) = X^T(n)A, \tag{6.31}$$

where $X^T(n) = [x(n)\, x(n-1) ... x(n-P)]$: data vector and $A = [a_0\, a_1, \ldots, a_P\,]$: filter coefficient vector.

If we assume the expected value of the signal to be zero, the variance of the output sequence can be written as

$$\begin{aligned} E(y(n)^2) &= E[a^{*T} X^*(n) X^t(n) a] \\ &= A^{*T} \Gamma_{XX} A, \end{aligned} \tag{6.32}$$

where Γ_{XX} is the autocorrelation matrix of $x(n)$. The filter coefficients are selected so that the response of the filter at every frequency is normalized to unity. If the variance of the output is now minimized [1], the filter passes that frequency, say, f_1, undistorted and attenuates other frequencies away from f_1. Changing f_1 over the entire range, we can find the power spectrum estimate with minimum variance.

6.4.2 Eigen Analysis Algorithm for Spectrum Estimation

Let us first discuss some concepts that are required to understand this method. Consider a square matrix of dimension $N \times N$ represented as A. The matrix A is said to have an eigenvector x and corresponding eigenvalue λ if

$$Ax = \lambda x. \tag{6.33}$$

Equation 6.33 holds good if

$$\det | Ax - \lambda x | = 0. \tag{6.34}$$

If Equation 6.34 is expanded, it will result in an Nth degree polynomial in λ. This polynomial is termed as the characteristic polynomial and the roots of the characteristic polynomial are called eigenvalues. The concepts are illustrated with a simple example.

Example 1

Find all the eigenvectors and eigenvalues of the matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

Solution: Let us first find the characteristic equation.

$$\det | Ax - \lambda I | = 0.$$

$$\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} = 0. \tag{6.35}$$

$$\lambda^2 - 4\lambda + 3 = 0 \Rightarrow (\lambda - 1)(\lambda - 3) = 0.$$

Eigenvalues are $\lambda = 1$ and $\lambda = 3$.

The eigenvector equation reduces to

$$| Ax - \lambda I | x = 0.$$

$$\begin{bmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0.$$

$$2x_1 + x_2 = \lambda x_1 \text{ and } x_1 + 2x_2 = \lambda x_2. \tag{6.36}$$

$$\text{Put } \lambda = 1 \Rightarrow x_1 + x_2 = 0 \Rightarrow x_1 = -x_2.$$

$$\text{The eigenvectors are} \begin{bmatrix} -x_2 \\ x_2 \end{bmatrix}.$$

$$\text{Put } \lambda = 3 \Rightarrow -x_1 + x_2 = 0 \Rightarrow x_1 = x_2.$$

$$\text{The eigenvectors are} \begin{bmatrix} x_2 \\ x_2 \end{bmatrix}. \tag{6.37}$$

Let us now try to understand the eigen analysis algorithm for power spectrum estimation. It was indicated previously that if the signal is generated using AR model and the signal has additive noise, the overall system becomes ARMA model. Let us consider a special case of the signal, for example, the sinusoidal signal corrupted by additive white noise. The overall system will again be ARMA model. To generate a sinusoid, let us use a difference equation for a system

$$x(n) - = -a_1 x(n-1) - a_2 x(n-2),$$

where $a_1 = 2\cos(2\pi f_k)$, $a_2 = 1$ with initial conditions

$$x(-1) = -1, x(-2) = 0. \tag{6.38}$$

The system has complex conjugate poles at f_k and $-f_k$.

The solution of the system is $x(n) = \cos(2\pi f_k n)$ for $n \geq 0$.

The generalized equation for generating P sinusoidal components can be written as

$$x(n) = -\sum_{m=1}^{2P} a_m x(n-m) \cdot \tag{6.39}$$

The transfer function of the system can be written as

$$H(Z) = \frac{1}{1 + \sum_{m=1}^{2P} a_m Z^{-m}}. \tag{6.40}$$

Let $A(Z) = 1 + \sum_{m=1}^{2P} a_m Z^{-m}$. It has $2P$ roots on the unit circle.

The roots of the denominator polynomial correspond to the frequencies of the sinusoids. Let the sinusoids be corrupted by white noise sequence, $w(n)$ with $E[|w(n)|^2] = \sigma_w^2$.

The signal with a number of sinusoids and additive white noise is represented as $y(n) = x(n) + w(n)$.

We can write x(n) as

$$x(n) - y(n) - w(n) = -\sum_{m=1}^{2P} [y(n-m) - w(n-m)]a_m \cdot \tag{6.41}$$

Equivalently, we can write

$$\sum_{m=0}^{2P} a_m y(n-m) = \sum_{m=0}^{2P} a_m w(n-m) \text{ with } a_0 = 1. \tag{6.42}$$

We can easily observe that Equation 6.42 represents the difference equation for ARMA model. The coefficients of the AR and MA system are identical. This is found to be the characteristic feature of the ARMA system for sinusoids with white noise.

The difference equation in Equation 6.42 may be expressed in the matrix form as

$$Y^T A = W^T A,\ A = [a_0, a_1, ..., a_m]. \tag{6.43}$$

If we multiply Equation 6.43 by Y and take the expected value, we obtain

$$\begin{aligned} &E[YY^T A] = E[YW^T A] = E[(X+W)W^T A]. \\ &\Gamma_{YY} A = \sigma_w^2 A. \\ &(\Gamma_{YY} - \sigma_w^2 I)A = 0. \\ &w(n) \text{ has zero mean and } X \text{ is a deterministic sequence.} \end{aligned} \tag{6.44}$$

Equation 6.44 can be recognized as an eigen equation with σ_w^2 as the eigenvalue of the autocorrelation matrix Γ_{YY}. We will understand Pisarenko harmonic decomposition method with the help of the following simple example.

Example 2

Autocorrelation values $\gamma_{YY}(0) = 3$, $\gamma_{YY}(1) = 1$, and $\gamma_{YY}(2) = 0$ are given for a process of a single sinusoid with additive white noise. Find the frequency, its power, and variance of white noise.

Solution: Let us first write the correlation matrix.

$$\Gamma_{YY} = \begin{bmatrix} \gamma_{YY}(0) & \gamma_{YY}(1) & \gamma_{YY}(2) \\ \gamma_{YY}(-1) & \gamma_{YY}(0) & \gamma_{YY}(1) \\ \gamma_{YY}(-2) & \gamma_{YY}(-1) & \gamma_{YY}(0) \end{bmatrix} = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 3 \end{bmatrix} \tag{6.45}$$

$$\gamma_{YY}(-1) = \gamma_{YY}(1) \text{ and } \gamma_{YY}(-2) = \gamma_{YY}(2)$$

The minimum eigenvalue is obtained as a root of the characteristic polynomial.

$$g(\lambda) = \begin{bmatrix} 3-\lambda & 1 & 0 \\ 1 & 3-\lambda & 1 \\ 0 & 1 & 3-\lambda \end{bmatrix} = (3-\lambda)(\lambda^2 - 6\lambda + 7) = 0.$$

The eigenvalues are $\lambda_1 = 3$, $\lambda_2 = 3+\sqrt{2}$, $\lambda_3 = 3-\sqrt{2}$.

$$\sigma_w^2 = \lambda_{\min} = 3-\sqrt{2}. \tag{6.46}$$

The corresponding eigenvalue is nothing but the vector that satisfies Equation 6.44.

$$(\Gamma_{YY} - \sigma_w^2 I)A = 0.$$

$$\begin{bmatrix} \sqrt{2} & 1 & 0 \\ 1 & \sqrt{2} & 1 \\ 0 & 1 & \sqrt{2} \end{bmatrix} \begin{bmatrix} 1 \\ a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \Rightarrow$$

$$a_1 = -\sqrt{2} \text{ and } a_2 = 1. \tag{6.47}$$

We will use the values of the coefficients to form a Z polynomial and find the roots to get the frequency.

$$a_0 + a_1 Z^{-1} + a_2 Z^{-2} = 0.$$

$$1 - \sqrt{2} Z^{-1} + Z^{-2} = 0.$$

$$Z^2 - \sqrt{2} Z + 1 = 0.$$

$$Z_{1,2} = \frac{1}{\sqrt{2}} \pm j\frac{1}{\sqrt{2}}, |Z_1| = |Z_2| = 1. \tag{6.48}$$

The roots lie on a unit code

We will now find the frequency using

$$Z_1 = e^{j2\pi f_1} = \frac{1}{\sqrt{2}} \pm j\frac{1}{\sqrt{2}} \Rightarrow f_1 = \frac{1}{8}. \tag{6.49}$$

The power of the sinusoid is obtained using the equation

$$P_1 \cos(2\pi f_1) = \gamma_{YY}(1) = 1 \Rightarrow$$

$$P_1 \times \frac{1}{\sqrt{2}} = 1 \Rightarrow P_1 = \sqrt{2} \quad (f_1 = 1/8, \cos(\pi/4) = 1/\sqrt{2}). \tag{6.50}$$

Concept Check

- What is the principle of minimum variance spectrum estimation?
- Define eigenvectors and eigenvalues.
- What is a characteristic equation?
- Write a difference equation to generate a sinusoid.
- State the characteristic feature of an ARMA system for sinusoids with white noise.
- How will you find the variance of noise from the eigenvalues?

6.5 Evaluation of Formants Using the Power Spectral Density Estimate

Let us discuss one application of power spectrum estimation, namely, evaluation of formants for a speech signal. But first let us understand the meaning of formants and their relation to LPCs using the vocal tract as an example.

The vocal tract can be considered as a circular waveguide with proper terminating conditions. It is closed at the larynx end and open at the mouth end. Excitation for voiced and unvoiced sound differs. It is a periodic pulse train for voiced speech generation. For unvoiced signals, excitation comes from the random noise generator or turbulence. When the excitation passes through the vocal tract, it resonates at certain resonating frequencies decided by the mode of excitation in the cavity, that is, TE10, TE20, TM11, mode, etc. These resonating frequencies are called formants.

Using these resonating frequencies, one can write the transfer function of the vocal tract, say in Z domain, as shown in Equation 6.51.

$$H(Z) = \frac{Z^M}{(Z - P_1)(Z - P_2),\ldots,(Z - P_M)}. \tag{6.51}$$

Here, we represent, say, M number of poles of the transfer function. This model of the vocal tract is called an all-pole model or AR model. In this model, it is assumed that there are zeros only at $Z = 0$, that is, the origin. If we multiply the factors of the denominator, we can obtain the denominator polynomial

$$H(Z) = \frac{Z^M}{Z^M + a_1 Z^{M-1} + a_2 Z^{M-2} + \cdots + a_{M-1} Z + a_M}, \tag{6.52}$$

where a_1, a_2, etc. are the coefficients of the polynomial called LPCs (as discussed in Chapter 5). If we factorize the denominator polynomial again, we will get the poles called formant frequency locations. This defines the relation between the formants and LPCs.

As we have observed, there are different methods to estimate power spectrum density. We have discussed the power spectrum density estimation using the method of periodograms in Section 6.2.1 and using the Welch method in Section 6.2.3.

As the offset used in the calculation of autocorrelation increases, the variance of the power spectrum density estimate also increases. There is a need to reduce the variance. This is where the Welch method comes in. Here, the method of periodograms is modified to reduce the variance of the estimate. The method is also called method of modified periodograms. Basically, this method divides the signal into small overlapping segments with 50% overlap between the successive segments. The second modification is that the data segments are first passed through a window, either Bartlett or Hamming, before calculation of the periodogram.

Similarly, for PSD estimates, the Welch method uses a window of small length, thereby decreasing the time window length and increasing the frequency window width, that is, decreasing the frequency resolution. The Heisenburg uncertainty principle states

$$\Delta t \times \Delta f \geq 1/2 \,. \tag{6.53}$$

When the time window length reduces, the variance of the estimate also reduces by the same factor. For example, if we are using a window of length 2048 for the periodogram method and a window of length 64 for the Welch method, the window length reduces by a factor of 2048/64, that is, 32. Hence, the variance of the estimate will reduce by a factor of 32. It must be remembered that improving the variance of the estimate is at the cost of frequency resolution. Some of the formants may be lost if the frequency resolution is not in a position to detect it.

We will now write a MATLAB program to estimate the power spectrum using the method of periodograms and the Welch method. We will use a direct command available in MATLAB, namely, spectrum.welch and spectrum.periodogram. The program listing given here is for a voiced speech segment using the Welch method. Figure 6.13 shows a plot of the signal and its periodogram. If we change the command in the program specrum. welch to spectrum.periodogram, we will get the periodogram estimate. Figure 6.14 shows the plot of the signal and its periodogram estimate.

```
%to find formants from the voiced signal using power spectral
estimation of speech signal using Welch method.
clear all;
fp=fopen('watermark.wav');
fseek(fp,224000,-1);
a=fread(fp,1024);
a=a-128;
subplot(2,1,1);plot(a);title('plot of voiced part of a signal');
xlabel('sample no.');ylabel('amplitude');
%1024 samples of the voiced signal are displayed
subplot(2,1,2);
Fs = 22100;   t = 0:1/Fs:.296;
h=spectrum.welch
d = psd(h,a,'Fs',Fs);
%Calculate the power spectrum estimate using Welch method
plot(d);
```

6.5.1 Interpretation of the Results

If we compare the plots of the power spectrum using the Welch method and the periodogram method, we find that the periodogram estimate almost tallies with the log spectrum graph. The peaks tracked in the power spectrum using the periodogram method tally with that obtained using log spectrum. The peaks occur at about 800, 1200, 1900, 3000, and

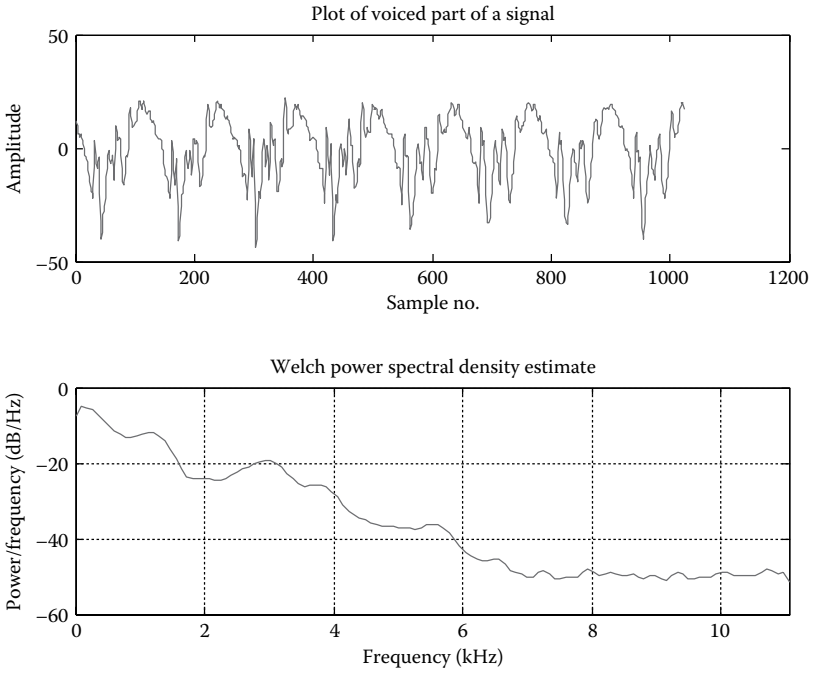


FIGURE 6.13
Plot of signal and its power spectrum estimate using the Welch method.

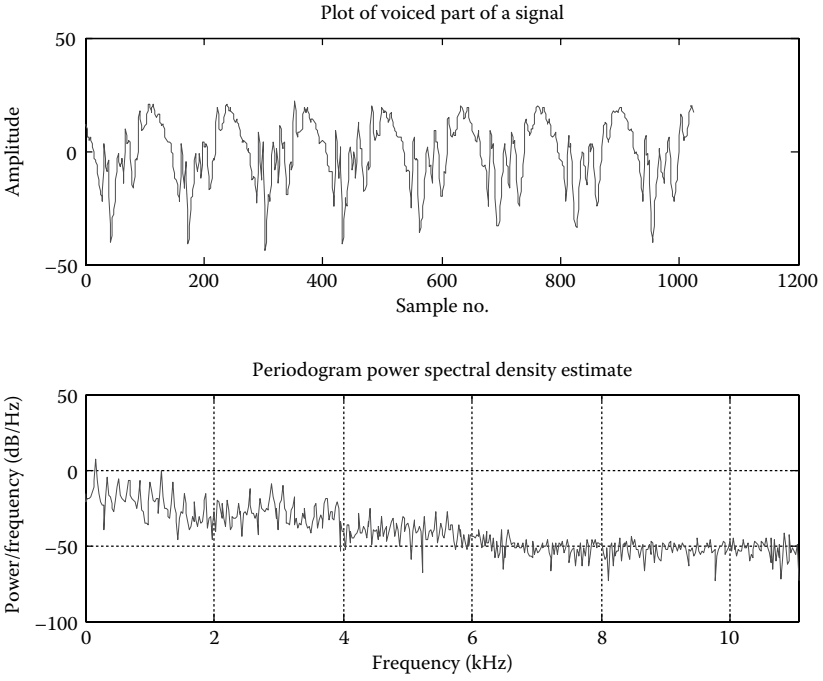


FIGURE 6.14
Plot of signal and its power spectrum estimate using the method of periodiogram.

3800 Hz. The peaks obtained using the Welch method are less in number; several peaks in the periodogram are missed when we use the Welch method. The Welch method used in the program uses a segment size of 64 allowing a frequency resolution of 22100/64, that is, 3431 Hz. The frequency points plotted are for 3431, 690.62, 10393, 1381.24, 1726.55, 2071.66, 2417.11, 2762.48, 3107.79 Hz, etc. The peaks for power spectrum density using the Welch method seem to occur at 1300, 3000, and 3800 Hz. The peaks at 800 and 1900 Hz are lost; we can say that the peaks near the frequency points resolved by the window are detected and other peaks are lost. To capture these peaks, one will have to use a longer data segment and a larger window size of, say, 2048 samples. If we do this, we will get almost the same power spectrum density graph as that for the periodogram method. We have to sacrifice for the variance of the estimate in this case. The reader is encouraged to verify this.

The same experiment can be repeated for the unvoiced segment of a signal. The program listing is given here for the Welch method. Figure 6.15 shows a plot of the signal and PSD estimate using the Welch method. Figure 6.16 shows the plot of the signal and its periodogram estimate for PSD.

```
%to find formants from the unvoiced signal using cepstrum of speech
signal
clear all;
fp=fopen('watermark.wav');
fseek(fp,23044,-1);
a=fread(fp,1024);
a=a-128;
subplot(2,1,1);plot(a);title('plot of unvoiced part of a signal');
xlabel('sample no.');ylabel('amplitude');
%1024 samples of the voiced signal are displayed
a1=window(@Hamming,1024);
for i=1:1024,
    a11(i)=a(i)*a1(i);
end
%the signal is passed via Hamming window and displayed.
Fs = 22100;   t = 0:1/Fs:.296;
h=spectrum.periodogram
d = psd(h,a11,'Fs',Fs);
figure;% Calculate the PSD
plot(d);
```

The observations made for the voiced segment can also be verified for the unvoiced segment. We see peaks in the PSD estimate using the Welch method at 1200, 2000, and 4000 Hz. The peaks using the periodogram method are found to occur at 800, 1200, 1500, 2200 Hz, etc. Hence, it is concluded that some of these peaks are lost using the Welch method.

Concept Check

- What are formants?
- What is the relation between formants and LPC?
- Why is it required to estimate the power spectrum density?

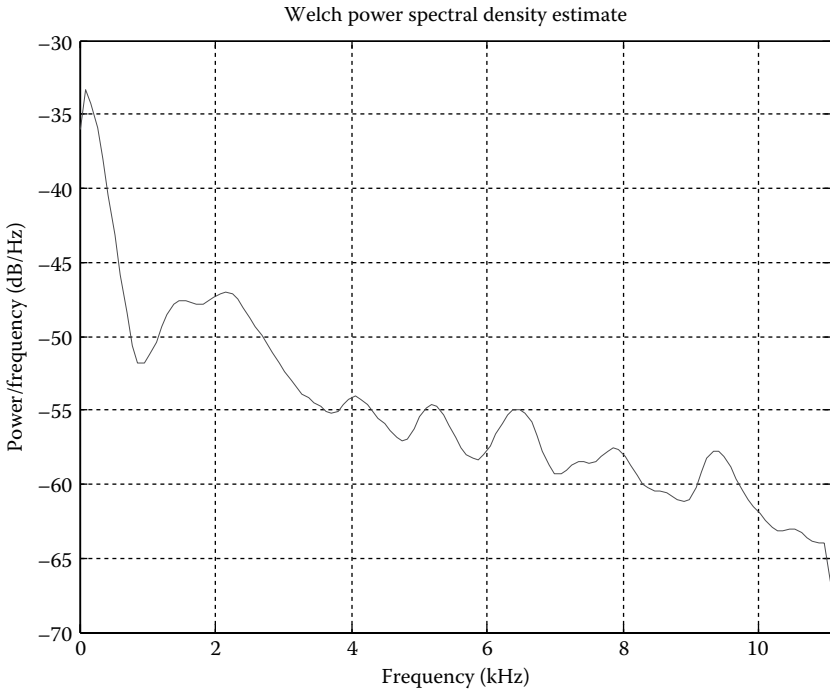


FIGURE 6.15
Plot of signal and its PSD estimate using Welch method.

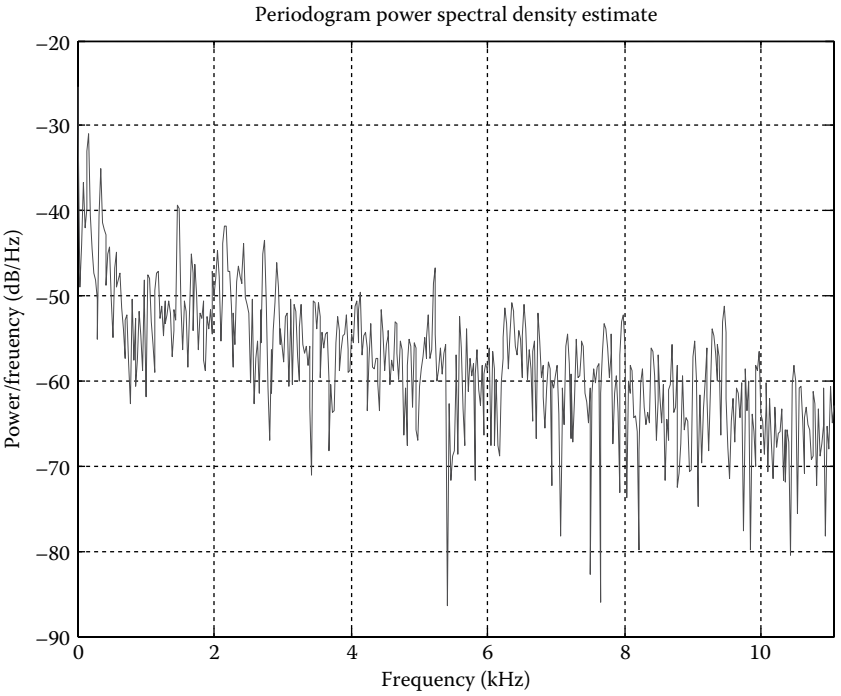


FIGURE 6.16
Periodogram power spectral density estimate.

- Why is an overlap of 50% recommended between the segments?
- Why do we call the Welch method a modified periodogram method?
- What is the drawback of the periodogram method?
- What is the drawback of the Welch method?

6.6 Evaluation of Cepstrum

This section is devoted to cepstral analysis. Let us first define a cepstrum. A cepstrum is obtained when we take an FT of the log spectrum. The term cepstrum is derived by reversing the first four letters of "spectrum." We can have a power cepstrum or a complex cepstrum.

Let us define a power cepstrum. It is a squared magnitude of the FT of the log of the squared magnitude of the FT of a signal. We can algorithmically define it as

$$\text{signal} \rightarrow \text{FT} \rightarrow \text{abs()} \rightarrow \text{square} \rightarrow \log \rightarrow \text{FT} \rightarrow \text{abs()} \rightarrow \text{square} \rightarrow \text{power cepstrum.}$$

Let us also define a complex cepstrum. The complex cepstrum was defined by Oppenheim in his development of the homomorphic system theory as the FT of the logarithm obtained from the unwrapped phase of the FT of a signal. Sometimes, it is also called the spectrum of a spectrum. We will define it algorithmically as

$$\text{signal} \rightarrow \text{FT} \rightarrow \text{abs()} \rightarrow \log \rightarrow \text{phase unwrapping} \rightarrow \text{FT} \rightarrow \text{cepstrum.}$$

The real cepstrum will use the log function defined for real values. The real cepstrum is related to the power cepstrum and the relation can be specified as

$$4(\text{Real spectrum})^2 = \text{Power spectrum.} \tag{6.54}$$

In some texts, the process of taking cepstrum is defined algorithmically as

$$\text{FT} \rightarrow \text{abs()} \rightarrow \log \rightarrow \text{IFT.}$$

This equation involves Inverse Fourier Transform (IFT) instead of FT. This hardly matters as when FT and IFT are applied on complex signals, the difference is only of a phase sign. The magnitude may change; but in this case, they define FT $X(k)$ and IFT $x(n)$ as given by Equations 6.54 and 6.55, respectively.

$$X(k) = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x(n) W^{nk}. \tag{6.55}$$

$$x(n) = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} X(k) W^{-nk}. \tag{6.56}$$

The algorithm takes FT of a signal; which means we enter the spectrum domain. Taking absolute value and log, we remain in the spectrum domain, that is, the *x*-axis is frequency. When we take inverse fast fourier transform (IFFT) further, we actually come back to the time domain. But here there is a log block in between which says that the output domain is neither a frequency domain nor a time domain. It is generally called a quefrency domain—quefrency obtained by reversing the first four letters of "frequency." Hence, a cepstrum graph of a signal will be an amplitude vs quefrency graph.

When a signal is analyzed in the cepstral domain, the analysis is termed as a cepstral analysis. A short-time cepstrum analysis was proposed by Schroeder and Noll for pitch determination of human speech.

Basically, speech signal is a convolved signal resulting from convolution of impulse response of the vocal tract with either a periodic pulse train or a random signal or turbulence for voiced and unvoiced speech, respectively. When one tries to find pitch frequency as a first peak in the spectrum, there is a possibility that the first formant and the fundamental frequency may overlap. In such a situation, it is difficult to separate the first formant from the pitch frequency. The isolation of these two components can be achieved in the cepstral domain. Hence, the cepstral domain is preferred for pitch measurement. Let us see how this is made possible in the next section.

6.7 Evaluation of Higher Order Spectra

Let us discuss higher order spectral analysis, namely, the bispectrum. We have studied power spectral estimation. It gives the distribution of power over the frequency components. Here, the phase information is suppressed. Power spectrum can be obtained as an FT of autocorrelation sequence. Hence, it gives the statistical information about a Gaussian process with its mean. Often, it is required to find information related to the deviation from the assumption of Gaussian processes. However, the process is nonlinear. Higher order spectra (HOS) helps in this regard. HOS are defined in terms of the higher order cumulants. Higher order moment spectra for deterministic signals are also defined. Higher order cumulant spectra are defined for random signals. The study of HOS is useful in three respects. First, it suppresses the Gaussian noise of unknown parameters. Second, it preserves the true phase and magnitude information. Third, it can detect nonlinearity in the signal.

We will first define moments, cumulants, and HOS of random signals. HOS are also called cumulant spectra and are defined in terms of the cumulants. Let us define cumulants. Consider a set of real random variables $x_1, x_2, ..., x_n$. Their joint cumulants of order $r = k_1 + k_2 + \cdots + k_n$ are defined as

$$C_{k_1 k_2, \ldots, k_n} = (-j)^r \frac{\partial \ln \varphi(\omega_1, \omega_2, \ldots, \omega_n)}{\partial \omega_1^{k_1} \partial \omega_2^{k_2}, \ldots, \partial \omega_n^{k_n}} \Big| \omega_1 = \omega_2 = \ldots = \omega_n = 0 \tag{6.57}$$

where $\varphi(\omega_1, \omega_2, ..., \omega_n) = E\{\exp[j\omega_1 x_1 + ... + \omega_n x_n]\}$ is their joint characteristic function.

The joint moments of the same set of random variables is given by

$$m_{k_1 k_2 \ldots k_n} = (-j)^r \frac{\partial \varphi(\omega_1, \omega_2, ..., \omega_n)}{\partial \omega_1^{k_1} \partial \omega_2^{k_2}, ..., \partial \omega_n^{k_n}} \Big| \omega_1 = \omega_2 = \cdots = \omega_n = 0$$

$$= E\{x_1^{k_1} x_2^{k_2} ... x_n^{k_n}\}. \tag{6.58}$$

The first four moments are given by

$$m_1 = E[x_1],\ m_2 = E[x_1^2],$$
$$m_3 = E[x_1^3],\ m_4 = E[x_1^4]. \tag{6.59}$$

The cumulants can be written in terms of moments as

$$C_1 = m_1,\ C_2 = m_2 - m_1^2,$$
$$C_3 = m_3 - 3m_2m_1 + 2m_1^2,$$
$$C_4 = m_4 - 4m_3m_1 - 3m_2^2 + 12m_2m_1^2 - 6m_1^4. \tag{6.60}$$

The computation of the *n*th cumulant requires the knowledge of all its previous moments. If the process is zero mean, that is, $m_1^x = 0$, then the second and third order cumulants are the same as their moments by putting $m_1^x = 0$ in Equation 6.60 for all *x*.

6.7.1 Cumulant Spectra

The HOS are called cumulant spectra as they are defined in terms of their cumulants. If the cumulant sequence is absolutely summable, then the nth order cumulant spectra of *x(k)* exists. The nth order cumulant spectra of *x(k)* is defined as the $(n-1)$ dimensional FT of the nth order cumulant sequence.

Case 1: Consider the power spectrum, that is, $n = 2$.

$$C_2^x(\omega) = \frac{1}{2\pi}\sum_{\tau=-x}^{x} c_2^x(\tau)\exp(-j\omega\tau),$$
$$\text{where } |\omega| \le \pi. \tag{6.61}$$

The second order spectrum is the FT of the autocorrelation sequence. This is the well-known result.

Case 2: Consider the trispectrum for $n = 3$. It is defined on similar lines.

$$C_3^x(\omega_1, \omega_2) = \frac{1}{(2\pi)^2}\sum_{\tau=-x}^{x} c_3^x(\tau_1, \tau_2)\exp[-j(\omega_1\tau_1 + \omega_2\tau_2)], \tag{6.62}$$

where $|\omega_1| \le \pi, |\omega_2| \le \pi, |\omega_1 + \omega_2| \le \pi$. There are two methods to find the HOS—the direct method and the indirect method. The definitions of cumulants depend on expectation evaluation that assumes the data to be of infinite length. In practice, the data available are of finite length. Hence, the cumulants can only be estimated. We will discuss the indirect method.

6.7.1.1 Indirect Method

Let $X(k)$ for $k = 1, \ldots, N$ be the available data. The algorithm can be stated as follows.

Step 1: Segment the data in K records of M samples each. Consider the ith segment or record of the data.

Step 2: Subtract the mean of each record from each sample of the record. This makes the data segment to have a zero mean. The third order cumulant will now be the same as the third order moment.

Step 3: Estimate the third order moment for each segment as follows.

$$m_3^{x_i}(\tau_1, \tau_2) = \frac{1}{M} \sum_{l=-l_1}^{l_2} x^i(l) x^i(l+\tau_1) x^i(l+\tau_2). \tag{6.63}$$

Step 4: Compute the average third order cumulant as

$$c_3^{x_i}(\tau_1, \tau_2) = \frac{1}{K} m_3^{x_i}(\tau_1, \tau_2). \tag{6.64}$$

Step 5: Obtain the bispectrum as

$$C_3^x(\omega_1, \omega_2) = \frac{1}{(2\pi)^2 K} \sum_{\tau=-x}^{x} m_3^x(\tau_1, \tau_2) \exp[-j(\omega_1 \tau_1 + \omega_2 \tau_2)] w(\tau_1, \tau_2),$$

$$\text{where } |\omega_1| \leq \pi, |\omega_2| \leq \pi, |\omega_1 + \omega_2| \leq \pi. \tag{6.65}$$

$w(\tau_1, \tau_2)$ is a two-dimensional window.

Let us do it practically. Let us use a speech signal and apply all the steps to find the bispectrum. A MATLAB program to do it is given here..

```
clear all;
f=fopen('watermark.wav','r');
fseek(f,80000,-1);
a=fread(f,2048);
plot(a);title('speech segment'); xlabel('sample no.');
ylabel('amplitude');
y = cum3est (a, 100, 128, 50, 'b', 32);
figure;
plot(y);title('third order cumulant estimated'); xlabel('sample no.');
ylabel('amplitude');
[Bspec,waxis] = bispeci (a, 100, 128, 50,'b',128, 0);
figure;
plot(abs(Bspec));title('bispectrum'); xlabel('sample no.');
ylabel('amplitude');
plot(angle(Bspec));title('phase of bispectrum'); xlabel('sample no.');
ylabel('angle in radians');
```

We have used the MATLAB commands, namely, cum3est and bispeci. The command cum3est estimates the third order cumulant for the data vector *a*. We have to specify the data vector *a*, the maximum lag to be computed (100), length of the data subsegments (128), and % overlap of the segments (50) have to be computed. The flag-if biased estimation is also to be computed; specify it as *b* or otherwise an unbiased estimation is done and the last parameter is fixed lag (32). The next command, namely, bispeci computes the bispectrum indirectly. It uses the parameters: data vector; maximum lag (100), length of the data subsegment (128), % overlap of the segments (50), which are all to be computed; flag-if biased estimation is also to be computed—specify it as *b* or otherwise unbiased estimation is done. The last parameter is nfft—FFT length is to be used [default = 128] and wind—window function is to be applied. If wind = 0, the Parzen window is applied (default), otherwise the hexagonal window with unity values is applied. Figure 6.17 shows a speech segment plot. Figure 6.18 depicts its third order cumulant and Figure 6.19 shows the plot of the bispectrum.

6.7.1.2 Direct Method

The algorithm for the direct method can be stated as follows.

Let $X(k)$ for $k = 1, \ldots, N$ be the available data. The algorithm can be stated as follows.

Step 1: Segment the data in K records of M samples each. Consider the ith segment or record of the data.

Step 2: Subtract the mean of each record from each sample of the record. This forces the data segment to have a zero mean. The third order cumulant will now be the same as the third order moment.

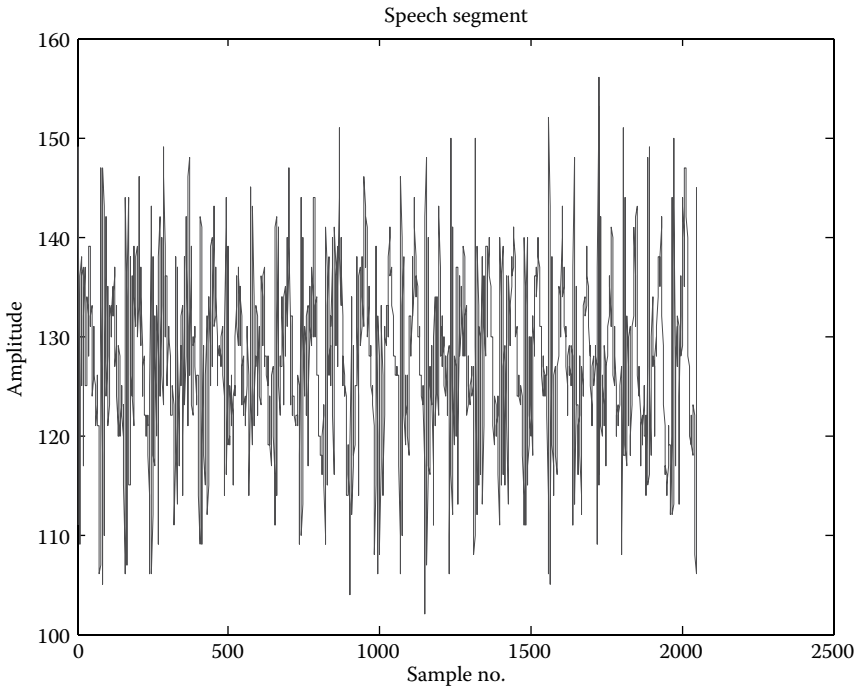


FIGURE 6.17
Plot of speech segment.

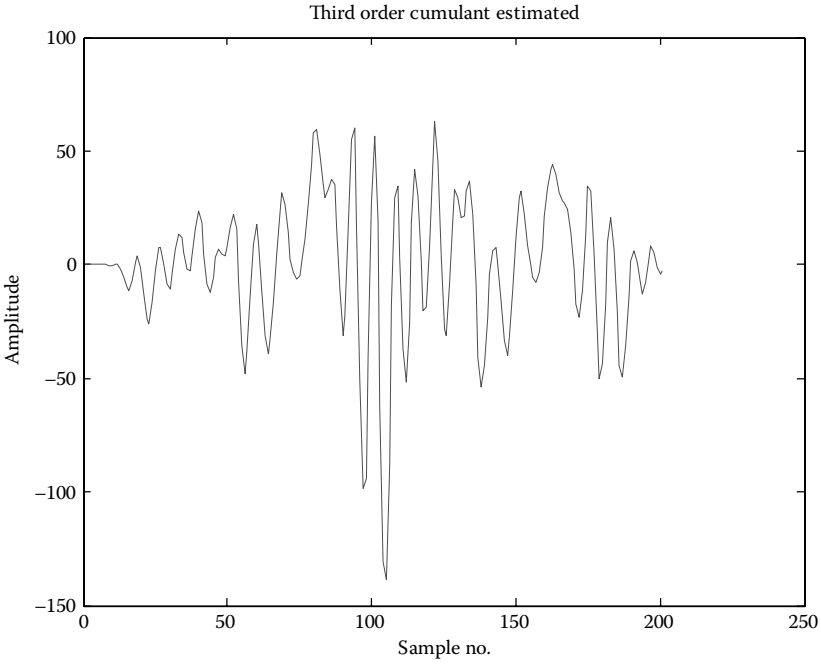


FIGURE 6.18
Third order cumulant estimated.

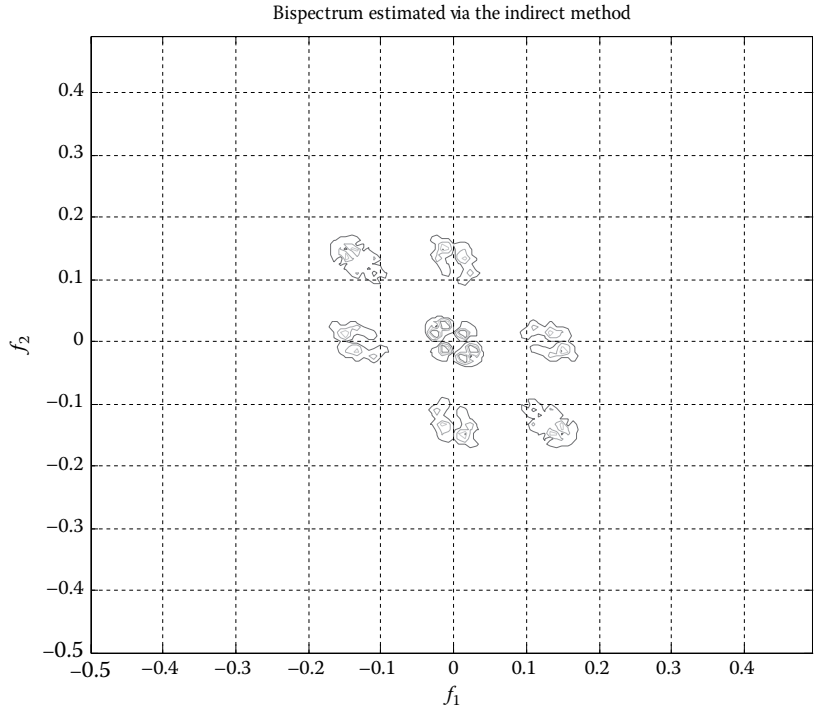


FIGURE 6.19
Plot of bispectrum.

Step 3: Find DFT of each segment.

$$F_X^i(k) = \sum_{n=0}^{M-1} x^i(n) \mathrm{e}^{-j\frac{2\pi}{M}nk}, \quad k = 0,1,\ldots,M-1, i = 1,2,\ldots,k. \tag{6.66}$$

Step 4: The third order spectrum of each segment is obtained as

$$C_3^{x_i}(k_{1,}, k_2) = \frac{1}{M} F_x^i(k_1) F_x^{i*}(k_2) F_x^{i*}(k_1 + k_2),\ i = 1,2,\ldots,K. \tag{6.67}$$

Step 5: The variance of the estimate can be reduced by smoothing over a rectangular window given by

$$C_3^{x_i}(k_{1,}, k_2) = \frac{1}{M_3^2} \sum_{n_1=-M_3/2}^{M_3/2-1} \sum_{n_2=-M_3/2}^{M_3/2-1} C_3^{x_i}(k_1 + n_1, k_2 + n_2),\ i = 1,2,\ldots,K. \tag{6.68}$$

Step 6: Finally, find the average of all third order spectra.

$$C_3^{x_i}(\omega_{1,}, \omega_2) = \frac{1}{K} \sum_{i=1}^{K} C_3^{x_i}(\omega_1, \omega_2),\ \omega_i = \frac{2\pi}{M} k_i, i = 1,2,\ldots,K. \tag{6.69}$$

A MATLAB program to find the bispectrum using direct method is given here.

```
clear all;
f=fopen('watermark.wav','r');
fseek(f,80000,-1);
a=fread(f,2048);
a=a-128;
plot(a);title('speech signal'); xlabel('sample no.');
ylabel('amplitude');
y = cum3est (a, 100, 128, 50, 'b', 32);
figure;
plot(y);title('third order cumulant estimated'); xlabel('sample no.');
ylabel('amplitude');
[Bspec1,waxis1] = bispecd (a,  128, 0, 128, 50);
figure;
subplot(2,1,1);
plot(abs(Bspec1));title('magnitude of bispectrum using direct
method'); xlabel('sample no.'); ylabel('amplitude');
subplot(2,1,2);plot(angle(Bspec1)); title('phase of bispectrum using
direct method'); xlabel('sample no.'); ylabel('amplitude');
```

Figure 6.20 shows the bispectrum plot using the direct method and Figure 6.21 shows a magnitude plot for the bispectrum for the same speech segment.

Let us find the bispectrum of a musical instrument signal, say, a flute. A MATLAB program to find the third order cumulant, bispectrum, bispectrum magnitude, and phase using the indirect method, and bispectrum, bispectrum magnitude, and phase using the direct method is given here.

Figure 6.22 shows a signal and its third order cumulant. Figure 6.23 plots the bispectrum using the indirect method.

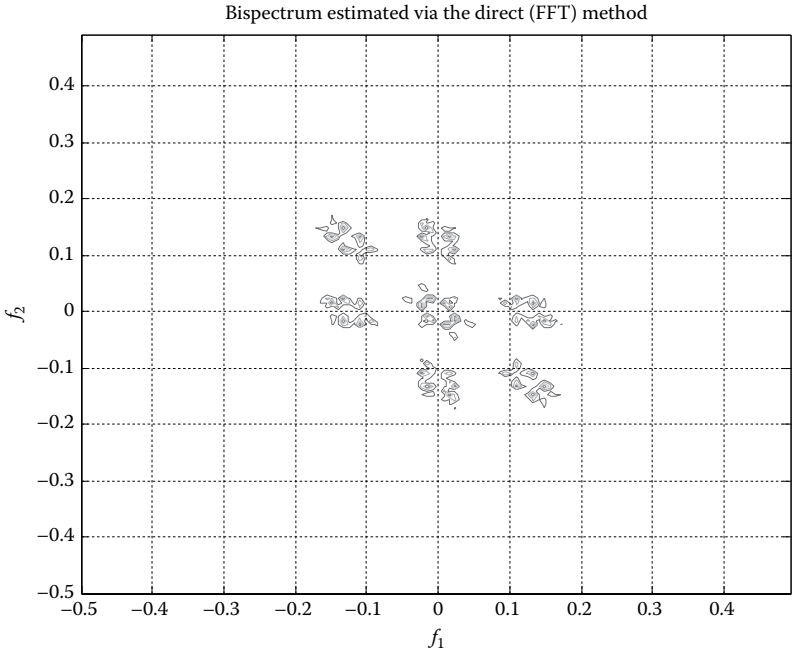


FIGURE 6.20
Bispectrum plot using the direct method.

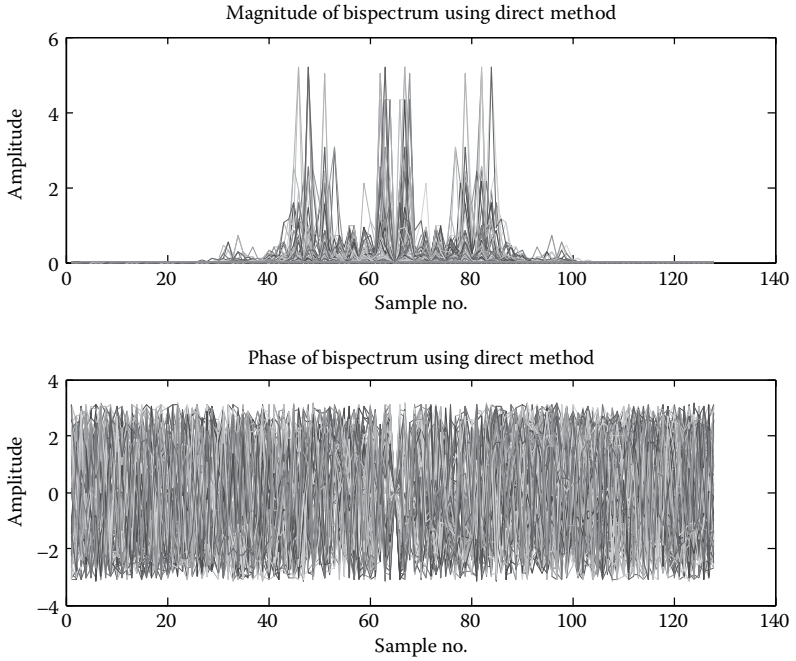


FIGURE 6.21
Magnitude and phase plot for a bispectrum using the direct method.

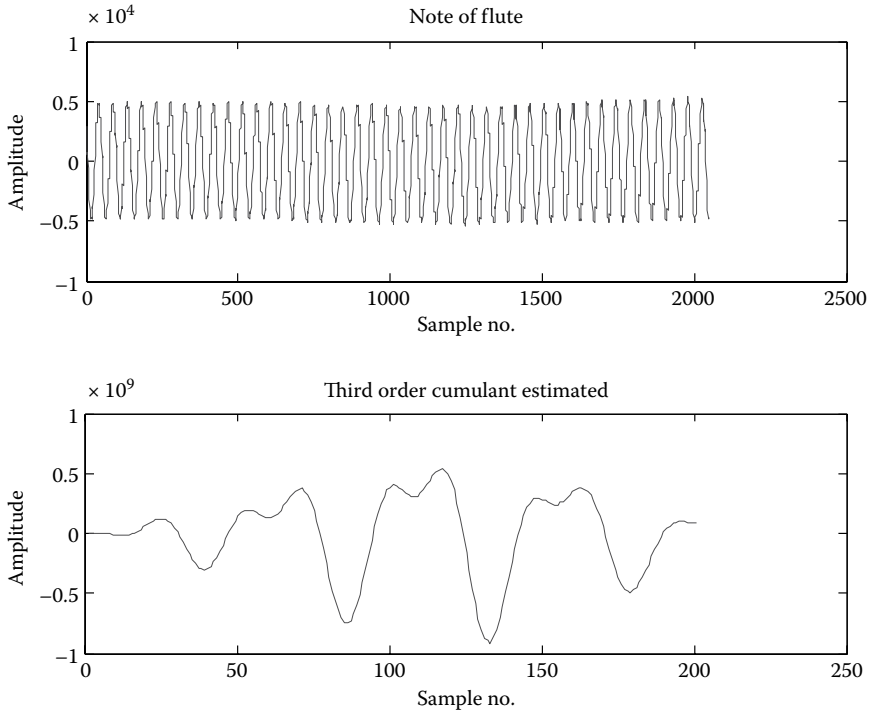


FIGURE 6.22
Plot of signal and its third order cumulant.

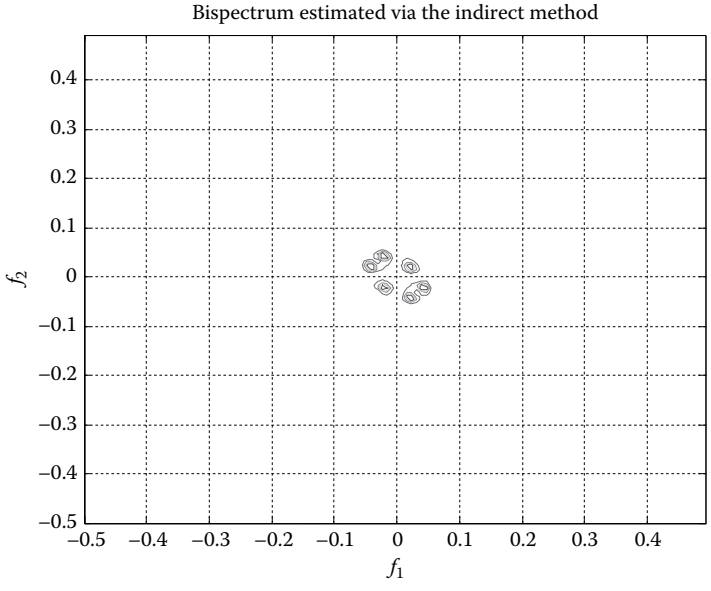


FIGURE 6.23
Plot of bispectrum using indirect method.

Trispectrum and further HOS are also computed on similar lines.

```
clear all;
%fp=fopen('Guitar_A_52.wav','r');% use this statement for Guitar bi
spectrum
fp=fopen('Flute_A_51.wav','r');
fseek(fp,90000,-1);
a=fread(fp,4096,'int16');
a=a-128;subplot(2,1,1);
plot(a);title('Note of flute');xlabel('Sample
number');ylabel('Amplitude');
y = cum3est (a, 100, 128, 50, 'b', 32);
subplot(2,1,2)
plot(y);title('third order cumulant estimated'); xlabel('sample no.');
ylabel('amplitude');
figure;
[Bspec,waxis] = bispeci (a,100,128,50,'b',128, 0);
figure;subplot(2,1,1);
plot(abs(Bspec));title('bispectrum magnitude using indirect method');
xlabel('sample no.'); ylabel('amplitude');
subplot(2,1,2);
plot(angle(Bspec));title('phase of bispectrum using indirect method');
xlabel('sample no.'); ylabel('angle in radians');
figure;
[Bspec1,waxis1] = bispecd (a,  128, 0, 128, 50);
figure;
subplot(2,1,1);
plot(abs(Bspec1));title('bispectrum magnitude using direct method');
xlabel('sample no.'); ylabel('amplitude');
subplot(2,1,2);
plot(angle(Bspec1));title('phase of bispectrum using direct method');
xlabel('sample no.'); ylabel('angle in radians');
```

Concept Check

- What are higher order cumulants?
- What are moments?
- What is the difference between moment and cumulant?
- How will you find the third order cumulant?
- Why do we use higher order spectra.
- Define a bispectrum.

Summary

This chapter mainly focuses on PSD estimation using nonparametric and parametric methods. Methods such as the minimum variance method and the eigen analysis method are also discussed. Lastly, HOS is introduced.

1. At first, energy signals and power signals are defined. Energy signals have finite energy and zero average power. Power signals have infinite energy and finite average power. The problems involved in calculation of energy density spectrum for deterministic signals are discussed. Energy density spectrum is calculated as the FT of autocorrelation. Power density spectrum calculation also has two problems, namely, the data record available is finite and the pdf is not known. We assume that the random process is ergodic so that ensemble autocorrelation is calculated as the time average autocorrelation.
2. Next, different nonparametric methods for power spectrum computation are discussed. Use of DFT to find the periodogram is explained. The Bartlett method uses average value of power for each DFT coefficient. Padding extra zeros to the signal extrapolates the periodogram. The Welch method is also termed as the method of modified periodograms. It allows the segments to overlap. The second modification suggested by Welch is to window the segments prior to computing the periodograms. The Blackman–Tukey method uses autocorrelation to find the power spectrum. It passes the autocorrelation via a smooth window to compensate for error in the calculation of the autocorrelation. The measures used for comparison of different nonparametric methods are quality and number of computations. The Welch method requires a little more computational power than the other methods. Quality is highest for the Blackman–Tukey method. MATLAB programs illustrate the different methods.
3. The next section concentrated on parametric methods using AR, ARMA, and MA model. The procedure for computation of power spectrum by finding AR parameters using the Levinson–Durbin algorithm is illustrated. Burg's method for power spectrum estimation is discussed. Burg uses forward and backward prediction error and minimizes the same in the algorithm. Burg postulated that the interpolation of autocorrelation that gives flattest spectrum will maximize the entropy. Hence, Burg's method is called as the MEM. The minimum variance method is described. If the variance of the output is minimized, the filter passes that frequency, say f_1, undistorted and attenuates other frequencies away from f_1. Changing f_1 over the entire range, we can find the power spectrum estimate with minimum variance. Eigenvalues and eigenvectors are defined. The eigen analysis method for power spectrum estimation is described. The sinusoidal signal corrupted by additive white noise is used. In this case, the overall system will again be an ARMA model. The coefficients of AR and MA system are identical.
4. One application of power spectrum estimation, namely, evaluation of formants in case of a speech signal, is described. We have briefly discussed the meaning of formants and its relation to LPC. We have illustrated the use of the periodogram method and the Welch method for evaluation of formants. The advantage of the Welch method and its drawback are discussed in detail with a MATLAB program.

5. Cepstrum domain is defined, and the pitch period measurement using cepstrum is illustrated using a block schematic. The method of entering a cepstrum domain is discussed, and the correct way to use a cepstrum window to eliminate the vocal tract information is pointed out. The peaks can be seen in the cepstrum in case of the voiced segment.
6. Finally, the higher order cumulant, namely, the third order cumulant, and HOS, that is, bispectrum, are discussed. Higher order cumulant spectra are defined for random signals. HOS suppresses the Gaussian noise of unknown parameters, preserves the true phase and magnitude information, and detects the nonlinearity in the signal. Higher order cumulants and spectra are defined. The indirect method and direct method for computation of bispectrum are explained. MALAB programs illustrate the computation of bispectrum for speech and music signals.

Key Terms

Energy signal
Power signal
Energy density spectrum
Power density spectrum
Periodogram
Modified periodogram
Bartlett method
Welch method
Blackman–Tukey method
AR model
ARMA model
MA model
Eigen analysis method
Minimum variance method
Discrete Fourier transform (DFT)
Autocorrelation
Formants
LPC
Time resolution
Frequency resolution
Heisenberg uncertainty principle
Levinson–Durbin algorithm
Berg's method
Maximum entropy method (MEM)
Lattice structure
Pitch period
Cepstrum
Cepstrum domain
Cumulants
Moments
Higher order spectra (HOS)
Bispectrum
Trispectrum
Magnitude plot
Phase plot

Multiple-Choice Questions

1. Energy signal is one for which
   a. Energy is highest.
   b. Energy is finite.
   c. Energy is infinite.
   d. Energy is finite and average power is zero.

2. Power signal is one for which
   a. Average power is finite and energy is infinite.
   b. Power is infinite.
   c. Power is zero.
   d. Power is infinite and energy is infinite.
3. Power density spectrum calculation is difficult because
   a. Data record is infinite.
   b. Finite data record is available.
   c. Probability density function is not known.
   d. Finite data record is available and probability density function is not known.
4. The Bartlett method of finding the periodogram is
   a. The method of averaging periodogram.
   b. The method of modified periodogram.
   c. The method of finding periodogram.
   d. The method of finding average of each DFT output.
5. Welch method is
   a. The method of averaging periodogram.
   b. The method of modified periodogram.
   c. The method of finding periodogram.
   d. The method of finding average of each DFT output.
6. Blackman–Tukey method
   a. Finds the average periodogram.
   b. Finds the periodogram.
   c. Finds autocorrelation and then FT of autocorrelation.
   d. Finds the autocorrelation.
7. Computational power required is more for
   a. Welch method.
   b. Blackman–Tukey method.
   c. Bartlett method.
   d. Periodogram method.
8. Parametric methods extrapolate
   a. The autocorrelation.
   b. The periodogram.
   c. The parameters.
   d. The Fourier transform
9. Levinson–Durbin algorithm
   a. Finds AR parameters.
   b. Finds MA parameters.

c. Finds ARMA parameters.
d. Finds lattice parameters.

10. The ARMA model is suitable when
    a. The signal is a clean signal.
    b. The signal is corrupted by additive noise.
    c. The signal is corrupted by multiplicative noise.
    d. The signal is corrupted by convoluting noise.
11. Matrix A is said to have an eigenvector x and eigenvalue λ if
    a. $Ax = \lambda$.
    b. $Ax = \lambda x$.
    c. $Ax = \lambda x^2$.
    d. $Ax = \lambda^2 x$.
12. The characteristic feature of ARMA system for sinusoids with white noise is that the
    a. Parameters of AR and MA model are the same.
    b. Parameters of AR model are different from MA model.
    c. Number of parameters of AR model is less than that of MA model.
    d. Number of parameters of AR model is more than that of MA model.
13. The formants are
    a. Resonating frequencies of vocal cords.
    b. Resonating frequencies of vocal tract.
    c. Resonating frequencies of both vocal tract and vocal cords.
    d. Resonating frequencies of mouth cavity.
14. The Welch method reduces the variance of the power spectrum estimate but
    a. Reduces frequency resolution.
    b. Reduces time resolution and frequency resolution.
    c. Increases frequency resolution.
    d. Increases time resolution.
15. The cepstrum is
    a. IFFT of the log spectrum.
    b. FFT of the spectrum.
    c. Windowing of the spectrum.
    d. Windowed FFT of the log spectrum.
16. If the speech segment is voiced,
    a. We observe repetition in the cepstrum.
    b. We observe peaks at regular intervals in the cepstrum.
    c. We observe the random nature in the cepstrum.
    d. We observe no peaks in the cepstrum.

17. Higher order cumulant spectra are defined for
    a. Deterministic signals.
    b. Random signals.
    c. Stationary signals.
    d. Strict sense stationary signals.
18. Higher order spectra are defined in terms of
    a. Cumulants.
    b. Moments.
    c. Autocorrelation.
    d. Fourier transform.
19. Burg postulated that the interpolation that gives flattest spectrum
    a. Will maximize the interpolation.
    b. Will maximize the entropy.
    c. Will maximize the error.
    d. Will maximize the spectrum.

Review Questions

1. Define energy signal and power signal. Define energy spectral density and power spectral density estimation. Why is it called an estimation? How can one estimate the pdf?
2. Describe estimation of power density spectrum of random signals using direct method and indirect method.
3. Describe the use of DFT for power spectrum estimation.
4. Draw a block schematic for the Bartlett method for power spectrum estimation and explain the algorithm.
5. Draw a block schematic for the Welch method for power spectrum estimation and explain the algorithm. What is the advantage of the Welch algorithm? Why is the Welch method also called the method of modified periodogram?
6. Compare the performance of different periodogram methods for power spectrum estimation. What are the performance measures?
7. Discuss the use of parametric methods for power spectrum estimation. Write the algorithm for estimation of power spectrum using the Yule–Walker method.
8. Describe Burg's method for power spectrum estimation.
9. Discuss the situation when Burg's method results in the maximum entropy method (MEM).
10. How will you use AR model for power spectrum estimation of voiced speech?
11. How will you use ARMA model for power spectrum estimation of signal corrupted with noise?

12. Explain minimum variance power spectrum estimation.
13. Explain the eigen analysis method for power spectrum estimation.
14. Define eigenvalues and eigenvectors. Explain the Pisarenko harmonic decomposition method with the help of simple example.
15. What are formants? How will you use periodogram for finding formants?
16. What is a cepstrum? How do we evaluate the cepstrum of a speech segment?
17. How will you take a V/UV decision from the cepstrum plot?
18. Define higher order cumulants. How will you find the bispectrum using indirect method?
19. How will you find the bispectrum using direct method?
20. What is the advantage of using higher order spectra?

Problems

1. Record a speech file of your voice and track the voiced speech. Read 1024 samples of speech and use DFT to compute the periodogram.
2. Record a speech file of your voice and track the voiced speech. Read 1024 samples of speech and use the Bartlett method to compute the periodogram. Compute the quality and number of complex multiplications.
3. Record a speech file of your voice and track the voiced speech. Read 1024 samples of speech and use the Welch method to compute the periodogram. Compute the quality and number of complex multiplications.
4. Record a speech file of your voice and track the voiced speech. Read 1024 samples of speech and use the Blackman–Tukey method to compute the periodogram. Compute the quality and number of complex multiplications.
5. Use the pyulear command in MATLAB to find the power spectrum for 1024 samples of speech file recorded of your voice. Use the Levinson–Durbin algorithm to find LPC.
6. Find all the eigenvectors and eigenvalues of the matrix $A = \begin{bmatrix} 4 & -1 \\ 1 & 2 \end{bmatrix}$.
7. The autocorrelation values
   $\gamma_{YY}(0) = 4$, $\gamma_{YY}(1) = 2$, and $\gamma_{YY}(2) = 0$ are given for a process of a single sinusoid with additive white noise. Find the frequency, its power, and the variance of white noise.
8. Find the first three formants for 1024 samples of speech recorded in your voice using the periodogram method and the Welch method.

Suggested Projects

1. Record a speech file in your own voice and use the Bartlett method to plot a periodogram for a segment of size 1024 samples.

2. Record a speech file in your own voice and use the Welch method to plot a periodogram for a segment of size 1024 samples.
3. Record a speech file in your own voice and use the Blackman–Tukey method to plot a periodogram for a segment of size 1024 samples.
4. Record a speech file in your own voice and use the AR model to plot a power spectrum for a segment of size 1024 samples.
5. Record a speech file in your own voice and use the AR model by Burg to plot a power spectrum for a segment of size 1024 samples.
6. Record a speech file in your own voice and use power spectrum estimation using the Welch method to find the formants for a segment of size 1024 samples.
7. Record a speech file in your own voice and use the power spectrum estimation using the Blackman–Tukey method to find the formants for a segment of size 1024 samples.
8. Record a speech file in your own voice and use cepstrum analysis to find the pitch period for a segment of size 1024 samples.
9. Record a speech file in your own voice and use the indirect method for higher order spectral analysis to find third order cumulant, a bispectrum, magnitude, and phase plot of bispectrum. Check if the bispectrum is different for different spoken words.
10. Record a musical instrument note and use the indirect and direct method for higher order spectral analysis to find third order cumulant, a bispectrum, magnitude, and phase plot of bispectrum. Check if the bispectrum is different for different instruments.

Answers

Multiple Choice Questions

1. (d)
2. (a)
3. (d)
4. (a)
5. (b)
6. (c)
7. (a)
8. (a)
9. (a)
10. (b)
11. (b)
12. (a)
13. (b)
14. (a)

15. (a)
16. (b)
17. (b)
18. (a)
19. (b)

Problems

1. Refer to Section 6.2.1. The only modification to be done is as follows. You have to record and plot the entire wav file and see by observation where the voiced part is. Use fseek command to track the voiced part and use the program given in the text on this voiced part.
2. Refer to Section 6.2.2 and Section 6.2. The only modification to be done is as follows. You have to record and plot the entire wav file and see by observation where the voiced part is. Use fseek command to track the voiced part and use the program given in the text on this voiced part.
3. Refer to Section 6.2.3 and Section 6.2. The only modification to be done is as follows. You have to record and plot the entire wav file and see by observation where the voiced part is. Use fseek command to track the voiced part and use the program given in the text on this voiced part.
4. Refer to Section 6.2.4 and Section 6.2. The only modification to be done is as follows. You have to record and plot the entire wav file and see by observation where the voiced part is. Use fseek command to track the voiced part and use the program given in the text on this voiced part.
5. Refer to Section 6.3.1. The only modification to be done is as follows. You have to record and plot the entire wav file and see by observation where the voiced part is. Use fseek command to track the voiced part and use the program given in the text on this voiced part.
6. Eigenvalues are $\lambda_1 = 3$ and $\lambda_2 =$. The eigenvectors are $\begin{bmatrix} x_2 \\ x_2 \end{bmatrix}$.

$$P_1 \cos(2\pi f_1) = \gamma_{YY}(1) = 2 \Rightarrow$$

7. $Z_1 = e^{j2\pi f_1} = -\frac{1}{2} \pm j\frac{\sqrt{3}}{2} \Rightarrow f_1 = \frac{1}{6}$. $P_1 \times \frac{1}{2} = 2 \Rightarrow P_1 = 4 \quad (f_1 = 1/6, \cos(\pi/3) = 1/2)$.

$$g(\lambda) = \begin{bmatrix} 4-\lambda & 2 & 0 \\ 2 & 4-\lambda & 2 \\ 0 & 2 & 4-\lambda \end{bmatrix} = (4-\lambda)(\lambda^2 - 8\lambda + 12) = 0.$$

The eigenvalues are $\lambda_1 = 4, \lambda_2 = 6, \lambda_3 = 2$.

$\sigma_w^2 = \lambda_{\min} = 2$.

8. Refer to Section 6. The only modification to be done is as follows. You have to record and plot the entire wav file and observe where the voiced part is. Use fseek command to track the voiced part and use the program given in the text on this voiced part.